System Level Modeling and Simulation of MVDC Microgrids featuring Solid State Transformers

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Applied Research in Power Electronics

Control of Power Converters

Modeling of Power Systems

- 2005-2009, EPFL-LEI CH, PhD. Degree
  SST and self-switching mechanisms, Prof. Alfred Rufer
- 2009, KTH-EME SE, Post Doc
  MMC sorting algorithms, Prof. Hans-Peter Nee
- 2010-2011, CERN CH, fellow researcher
  Powering strategies and reliability
- 2011-2012, ABB MV-Drives CH, control engineer
  Control of AC active front ends
- 2013-2013, CERN CH, fellow researcher
  High voltage resonant converter
- 2014-2016, HEIG-VD CH, research collaborator
  Power Electronics teaching activities
- 2014-2020 PESC-CH, independent consultant
  DC substation energy recovery, modular BESS solutions
- 2020, Hitachi Energy CH, senior R&D engineer
  SST, E-mobility, MMC Static Frequency Converters

Tutorial Outline

The tutorial will be composed half/half with traditional slides and practical demonstrations that the audience can follow and redo on their own computers.

The models are built and demonstrated with *Powersys Simba**, a simple and powerful simulation environment that allows Python automated scripting and a so-called continuous time simulation approach.

**the presented models are easily implementable in other similar simulation software environments*

- Introduction
  - DC grids basics, Standards
- System level Models
  - Elementary DAB-based SST cell
  - DC bus, active front end and batteries, Series/parallel connection of SSTs
- Model of an MVDC microgrid
  - DC bus with multiple AFE/PFEs, Full model of an MVDC power system
  - Continuous run approach
- Conclusions and Prospects

INTRODUCTION

References:

DC Distribution Systems and Microgrids, edited by Tomislav Dragičević, Pat Wheeler, Frede Blaabjerg, IET 2018

Medium Voltage DC system Architectures, edited by Brandon Grainger, Rik De Doncker, IET 2021

- DC grids basics
  - Grid Modelling, Standards
- Norton/Thevenin equivalent sources
  - grid and impedance, current/voltage sources
- Software for tutorial examples
  - Simba and Python 3.8

DC Grid Basics

- When compared to AC grids, similarities can be drawn:
  - Primary control is related to active power (no reactive), preferably implemented via droop control (also related to virtual resistance).
  - Secondary and Tertiary control are related to power and grid management, requiring communication between converter units or between grids.
  - A grid can be modelled with ideal sources and its Thevenin/Norton equivalent reflecting its impedance.

- When modelling DC grids for study there are still some open questions
  - Protection is not as simple as in AC grid because there is no zero current crossing for easy breaking features, requiring solid-state breakers.
  - What does short circuit capability mean in DC grids? Is it related to converters' limits while operating DC transformers?
  - What does inertia mean in DC grids? In AC grids, inertia is related to the rotating momentum of generators. Is DC inertia related to stored energy in capacitors?

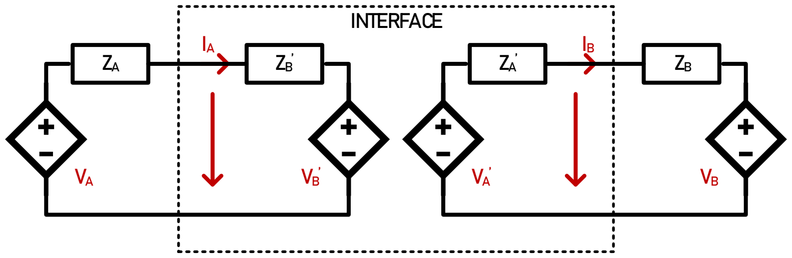
About Standards

- Lack of standardized protocols for components and systems integration
- some recommendations can be found
  - Cigre Technical Brochure: Medium Voltage DC Distribution Systems (TB875)
  - MDPI open access: DC Microgrids: Benefits, Architectures, Perspectives and Challenges
  - IEEE P3105: Recommended Practice for Design and Integration of Solid State Transformers in Electric Grid *(in work)*

- IEEE Std 1709-2018
  - This document is IEEE Recommended Practice for 1 kV to 35 kV MVDC power systems on ships.
    So far, it is the only standard governing MVDC applications
- IEEE 2030.10-2021
  - This document is IEEE Standard for LVDC Microgrids for Rural and Remote Electricity Access Applications.
- IEC TR 63282:2020
  - LVDC systems -Assessment of standard voltages and power quality requirements

Inputs from HIL-RTS

- The work behind the modeling that is presented in this tutorial is inspired by a common practice in Hardware-in-the-Loop Real Time Modelling where all systems and subsystems are defined through their Thevenin/Norton equivalent.



- IEEE P2004 (DRAFT) : Recommended Practice for Hardware-in-the-Loop (HIL) Simulation-Based Testing of Electric Power Apparatus and Controls
- This is interesting because it discusses the way to interface subsystems through equivalent source and equivalent impedence as seen from the other subsystem

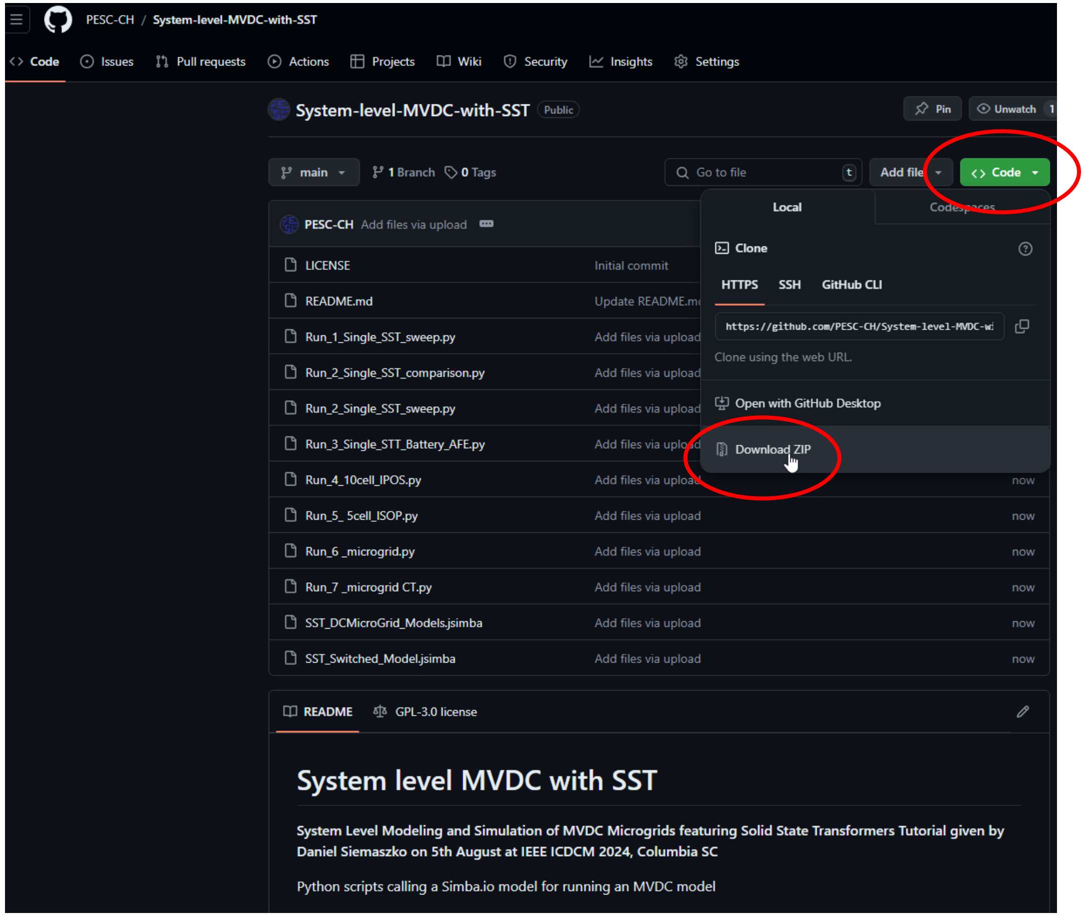
INSTALLING PYTHON 3.8

- [Python Release Python 3.8.0 | Python.org](https://www.python.org/downloads/release/python-380/)
  https://www.python.org/downloads/release/python-380/

- Then in terminal type:
  - pip install math
  - pip install numpy
  - pip install mathplotlib
  - pip install aesim.simba
  - pip install os-sys
  - pip install tk

GET THE MODELS

- https://github.com/PESC-CH/System-level-MVDC-with-SST and download zip archive



SYSTEM LEVEL MODELS

- Elementary DAB-based SST cell
  - definition, model, current/voltage control
- DC bus, Active Front End and batteries
  - Definitions and interactions
- Series/Parallel connection of SST cells
  - ISOP, IPOS elementary configurations

References:

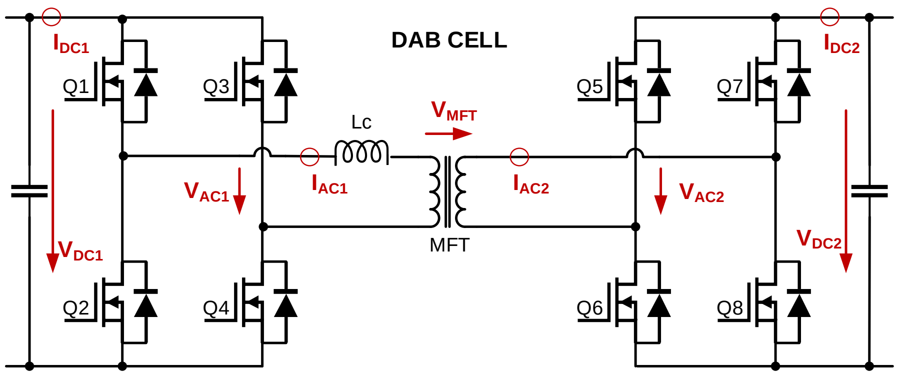
D. Siemaszko and P. Noisette, "**Power System Simulation Tool for Quick Benchmarking of Innovative MVDC Grids in E-Mobility Applications**," 2022 24th European Conference on Power Electronics and Applications (EPE'22 ECCE Europe), Hanover, Germany, 2022

S. Heinig and D. Siemaszko et al., "**Experimental Insights into the MW Range Dual Active Bridge with Silicon Carbide Devices**," 2022 International Power Electronics Conference (IPEC-Himeji 2022-ECCE Asia), Himeji, Japan, 2022

ELEMENTARY DAB-BASED SST CELL · DEFINITION

- Dual Active Bridge (DAB) is a promising candidate for acting as modular SST building bloc
- The active power flow is phase shift controlled through pulses applied to the Medium Frequency Transformer

- This study will implement a system level model of a converter that has been built and tested in laboratory
- The operating parameters and passive elements reflect its overall dynamics and behaviors

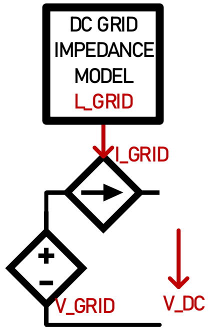


S. Heinig et D. Siemaszko et al., "Experimental Insights into the MW Range Dual Active Bridge with Silicon Carbide Devices," 2022 International Power Electronics Conference (IPEC-Himeji 2022- ECCE Asia), Himeji, Japan, 2022

	Symbol	Value	Comment
DC nominal voltage on primary side	V_{PRIM}	1000V	Set by grid model V_{GRID}
DC nominal voltage on secondary side	V_{SEC}	2000V	Current/Voltage control by SST
nominal current on secondary side	I_{SEC}	250A	Set by load model I_{LOAD}
maximum current on secondary side	I_{SEC_MAX}	300A	Voltage control Limiter by SST
MFT side SST inductance	L_{MFT}	8.5µH	Defines control's time response
DC side SST capacitor	C_{DC}	3mF	Defines control's time response
MFT switching frequency	F_{SW}	10kHz	Defines control's transfer function
Simulation time step	t_s	1µs	This time step is used in all runs

ELEMENTARY DAB-BASED SST CELL · MODEL

- The DAB based SST is modelled with it's equivalent Thevenin/Norton equivalents.
- Voltage and current sources are controlled in a way it reflects its dynamics as seen from its terminals.
- The grid model is an ideal DC voltage source with a current source in series that implements the DC grid impedance.
- The impedance model is given through its equivalent RL values.



$$I_{GRID} = \frac{1}{L_{GRID}} \int (V_{DC} - V_{GRID} - R_{GRID} I_{GRID})$$

ELEMENTARY DAB-BASED SST CELL · MODEL

- The MFT is modelled with two current sources (one per winding)
- The current is related to leakage inductor L_{MFT}, switching frequency F_{SW} and applied voltage V_{MOD}

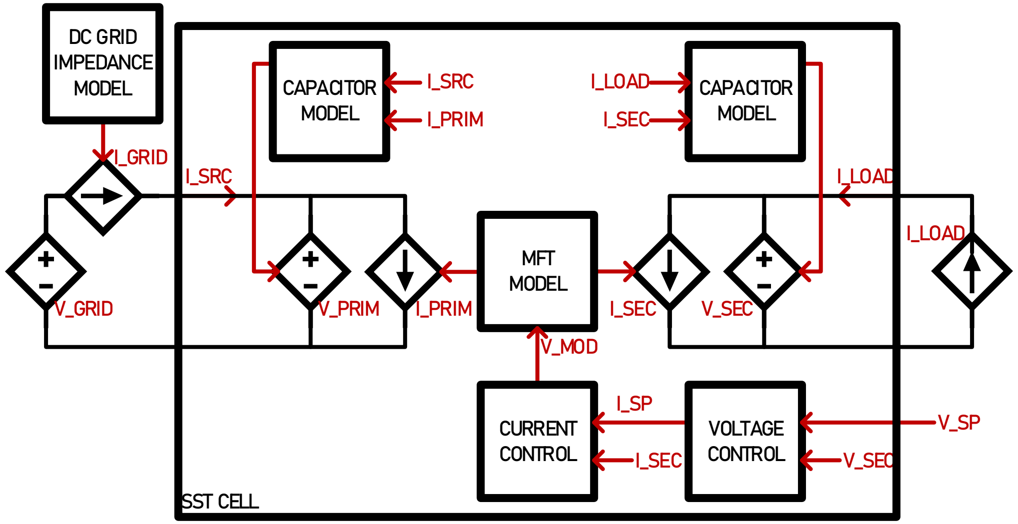
$$V_{PRIM} = \frac{1}{C_{DC}} \int (I_{SRC} - I_{PRIM})$$

$$V_{SEC} = \frac{1}{C_{DC}} \int (I_{LOAD} - I_{SEC})$$

$$I_{SEC} = \frac{1}{L_{MFT} F_{SW}} V_{MOD}$$

$$I_{PRIM} = I_{SEC} N_{MFT}$$

- Capacitors are modelled with voltage sources
- The voltage is related to difference in currents from interface and winding, and capacitor value C_{DC}

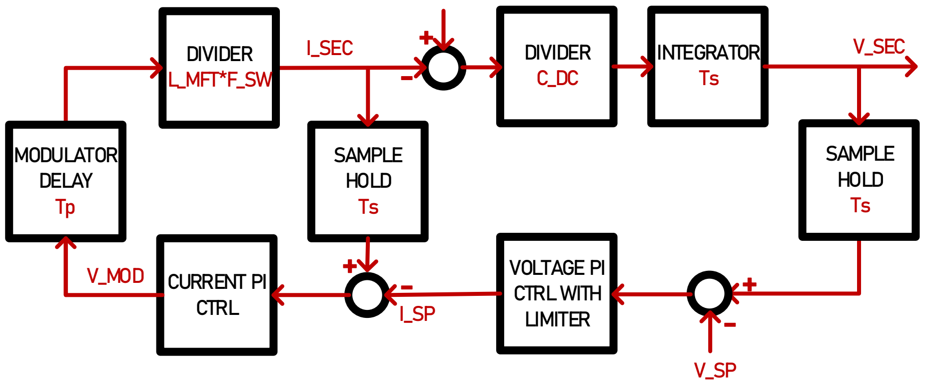


ELEMENTARY DAB-BASED SST CELL · MODEL

- The modulation voltage V_{MOD} applied to the leakage inductor is given by a current/voltage cascaded PI controller
- The full system implements delays related to sampling frequency T_S and modulator's inherent delay due to switching period T_P

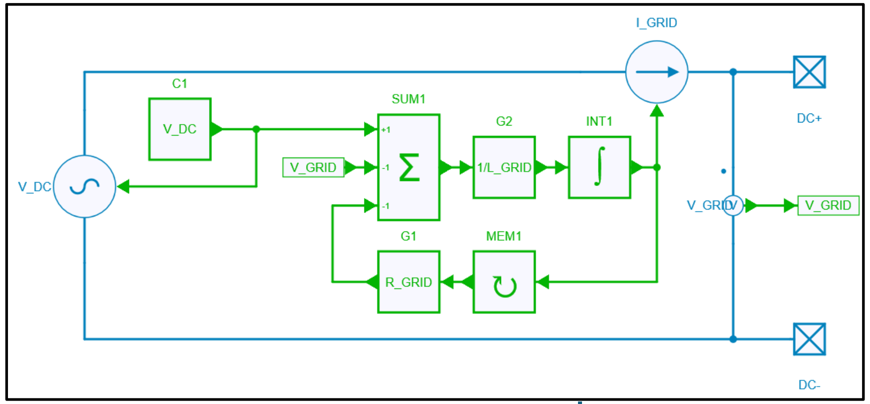
$$V_{MOD} = K_P(I_{SEC} - I_{SP}) + K_I \int (I_{SEC} - I_{SP})$$

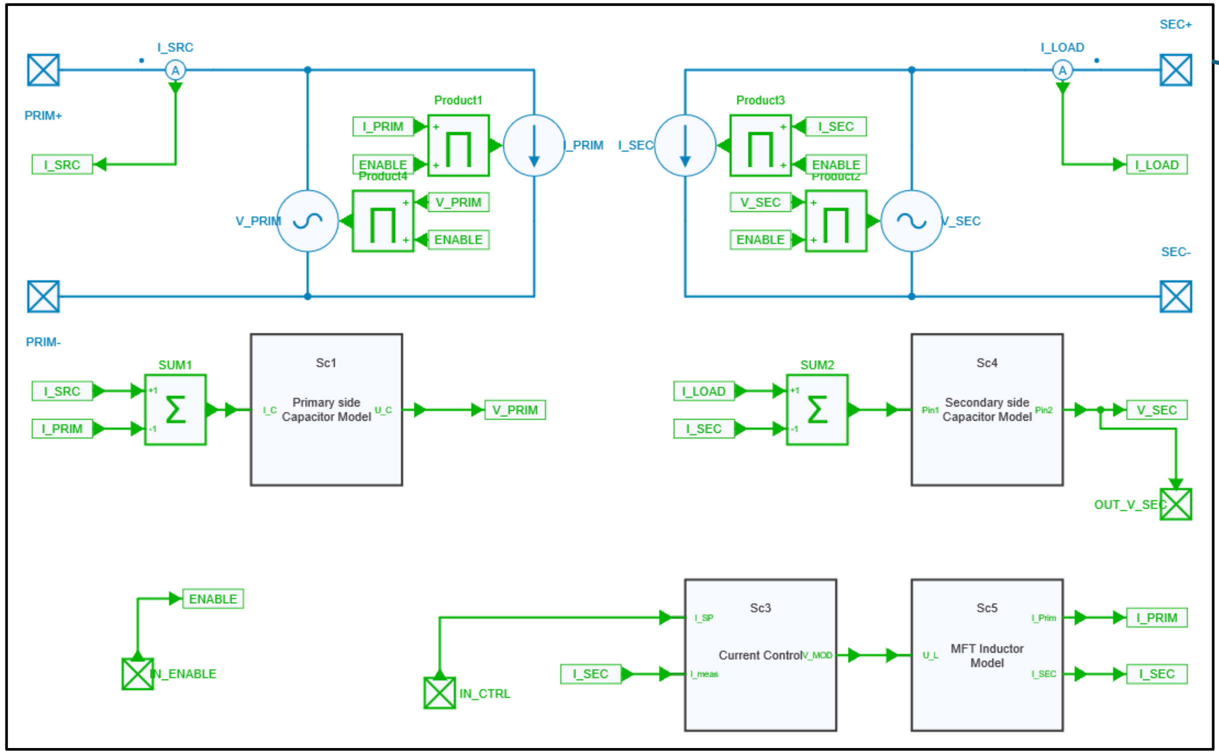
$$I_{SP} = K_P(V_{SEC} - V_{SP}) + K_I \int (V_{SEC} - V_{SP})$$

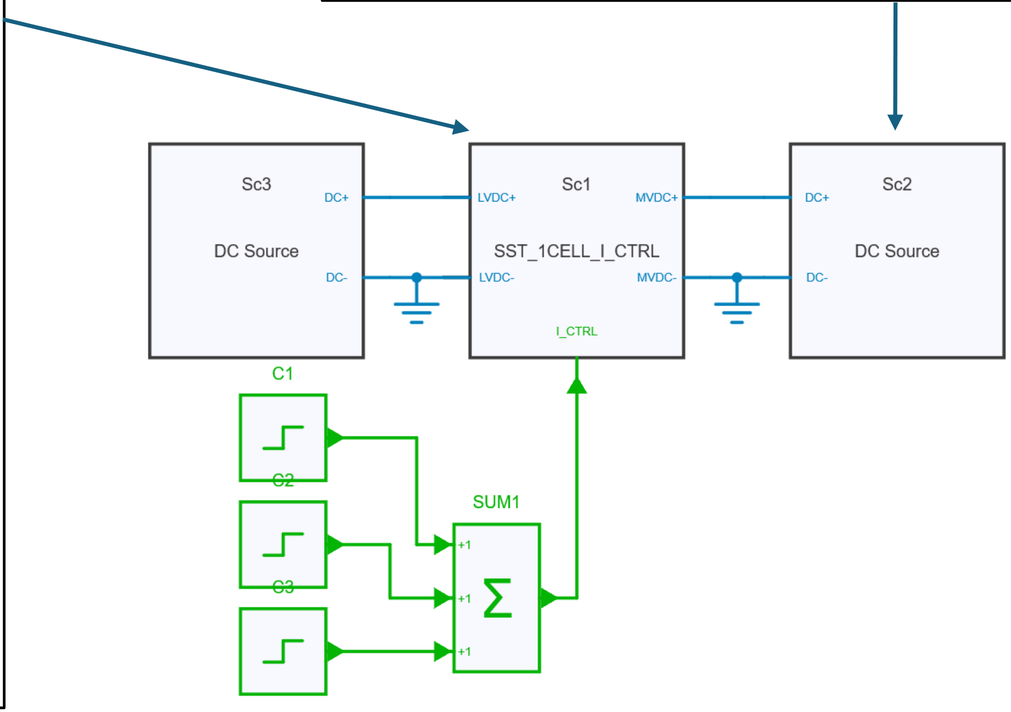


ELEMENTARY DAB-BASED SST CELL · CURRENT CONTROL

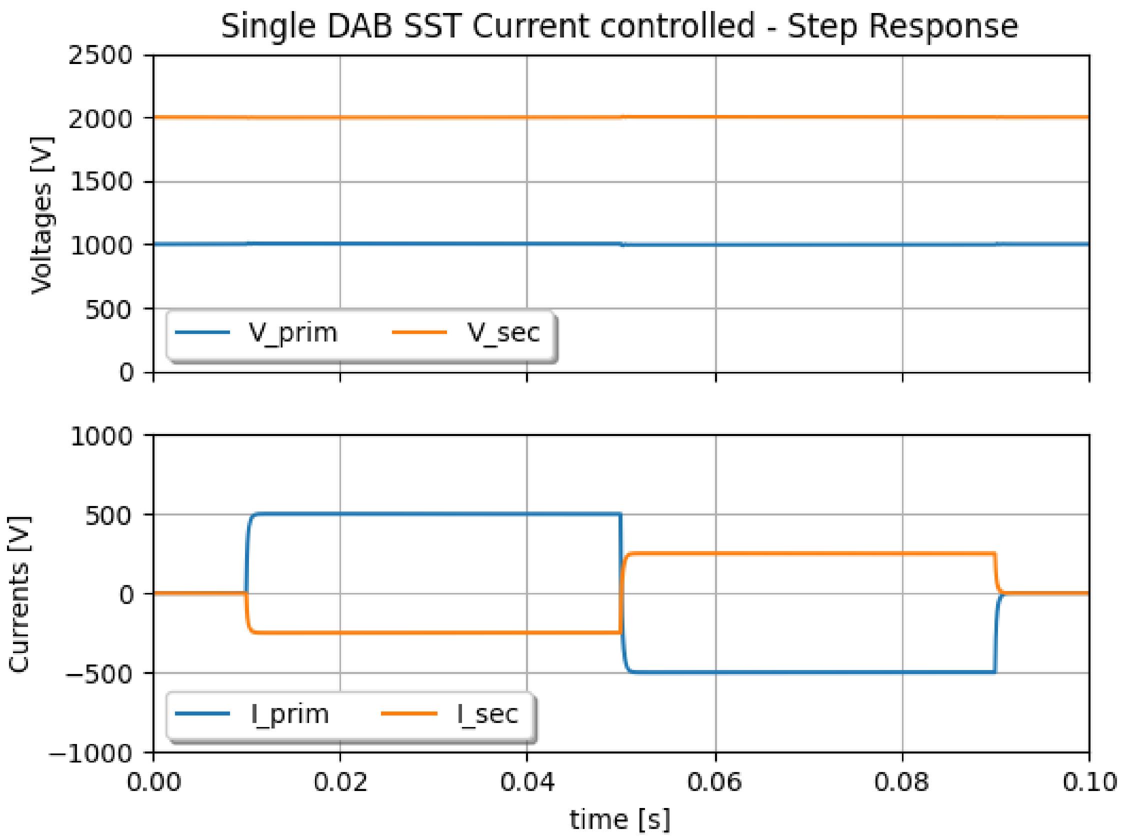
- *Run model "1 Single SST Current CTRL" from file "SST_DCMicroGrid_Models.jsimba" or Python script "Run_1_Single_SST_perturbation" and "Run_1_Single_SST_sweep"*

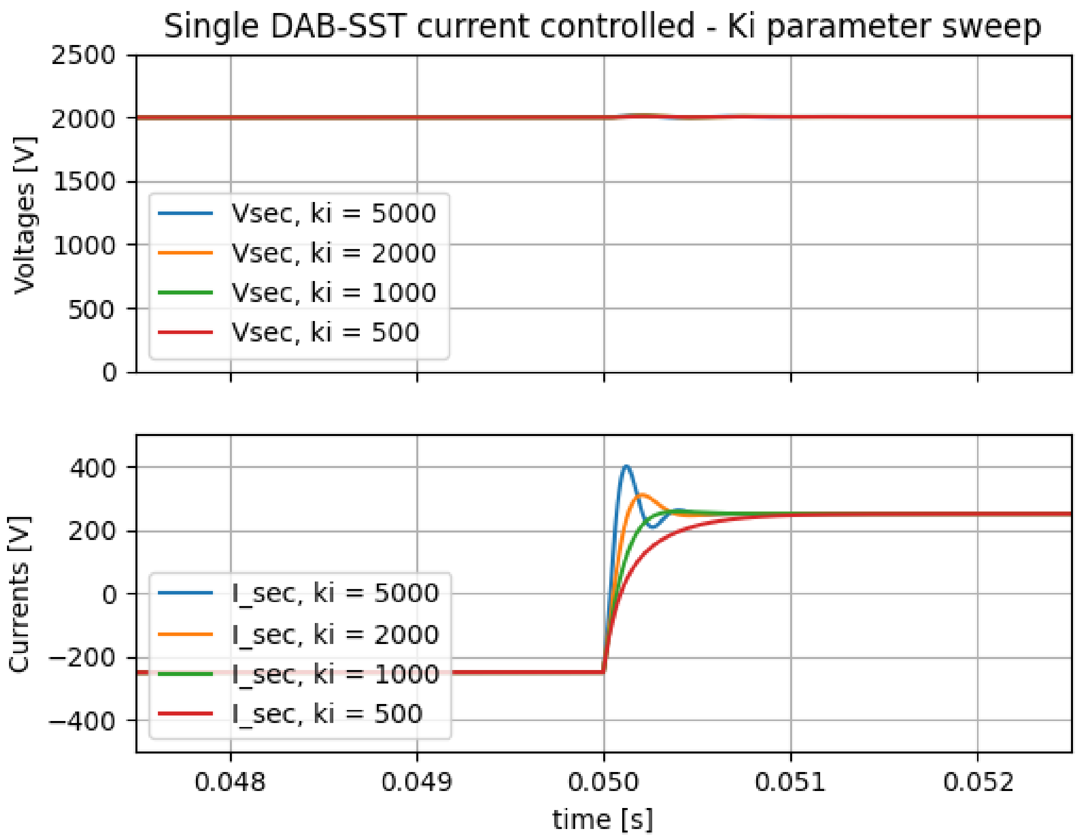






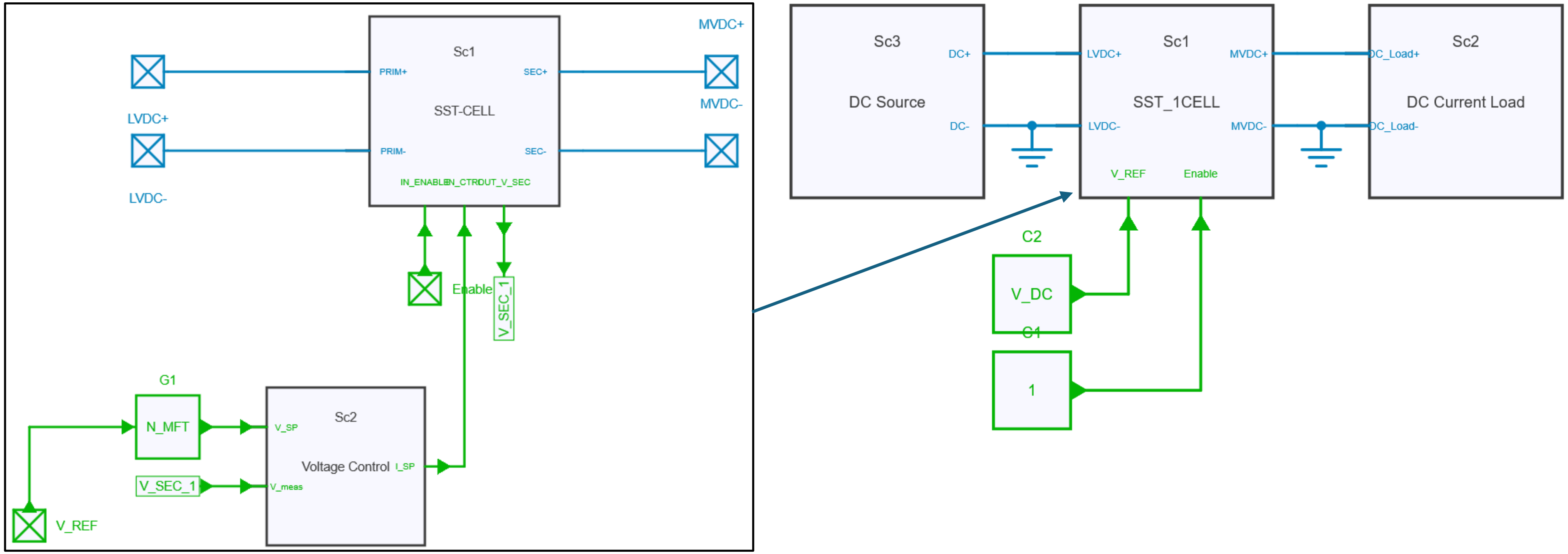
Elementary DAB-based SST cell · current control



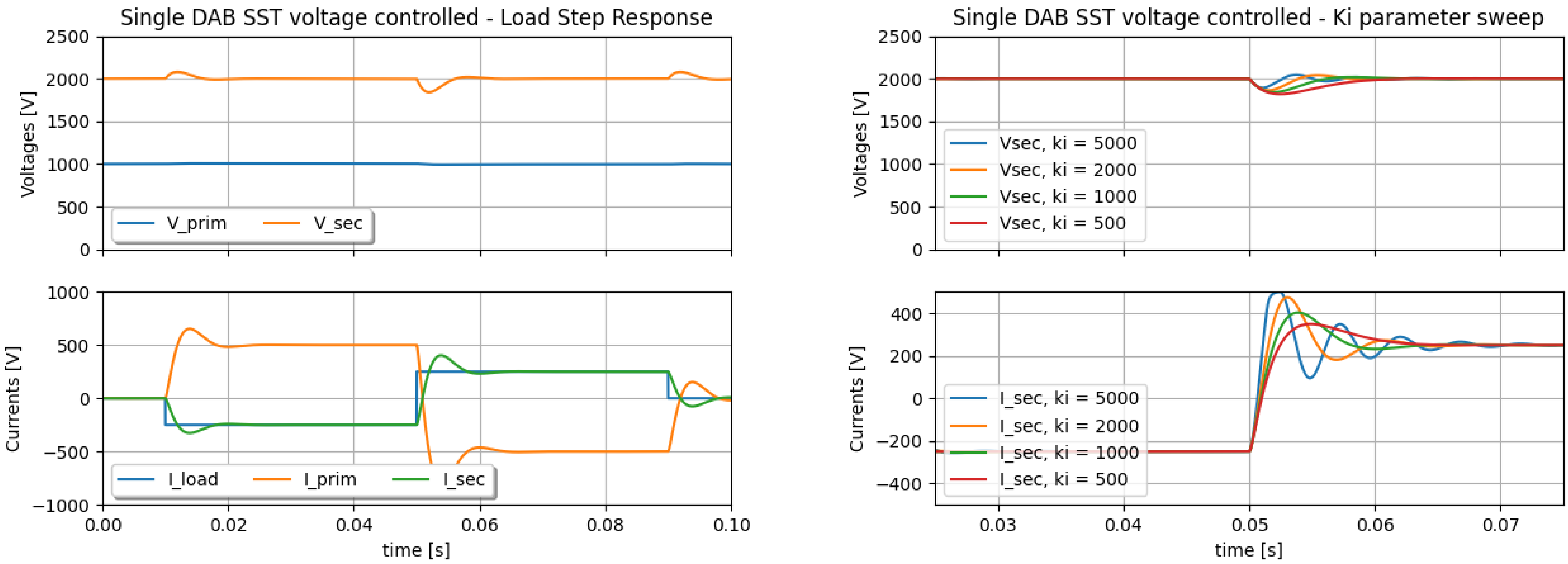


Elementary DAB-based SST cell · voltage control

- *Run model "2 Single SST"*
  from file "SST_DCMicroGrid_Models.jsimba"
  or Python script "Run_2_Single_SST_sweep"

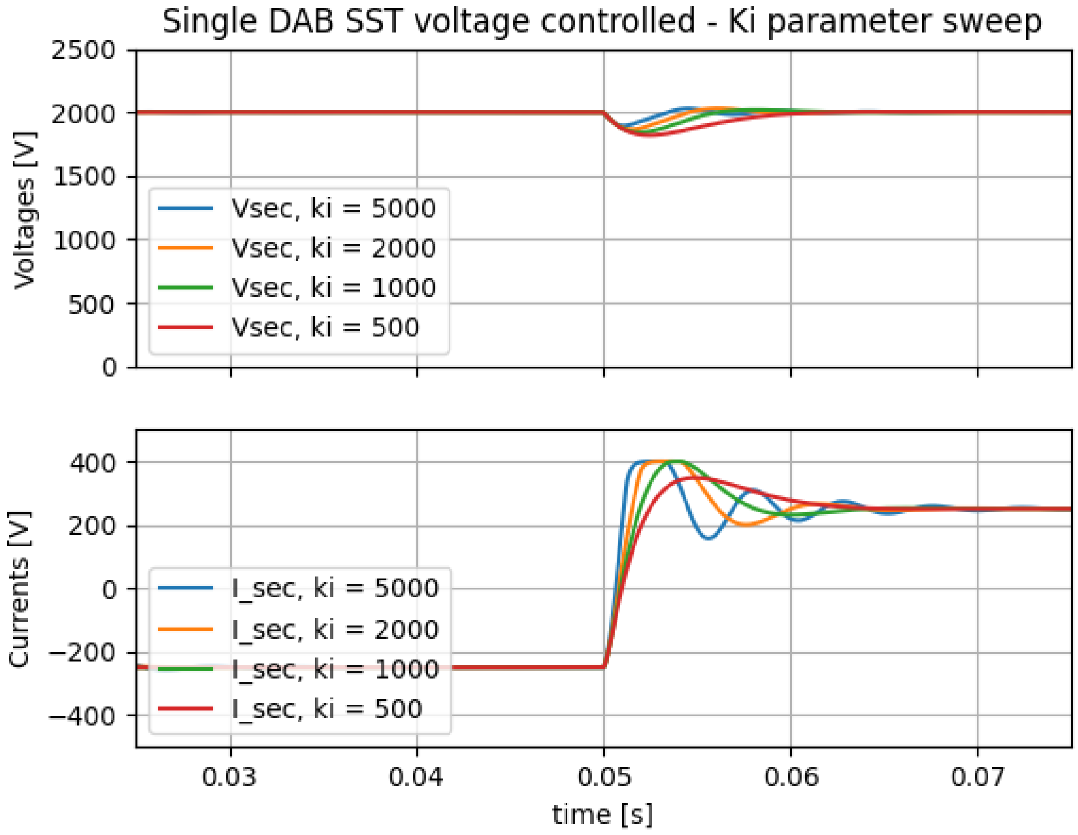


ELEMENTARY DAB-BASED SST CELL · VOLTAGE CONTROL



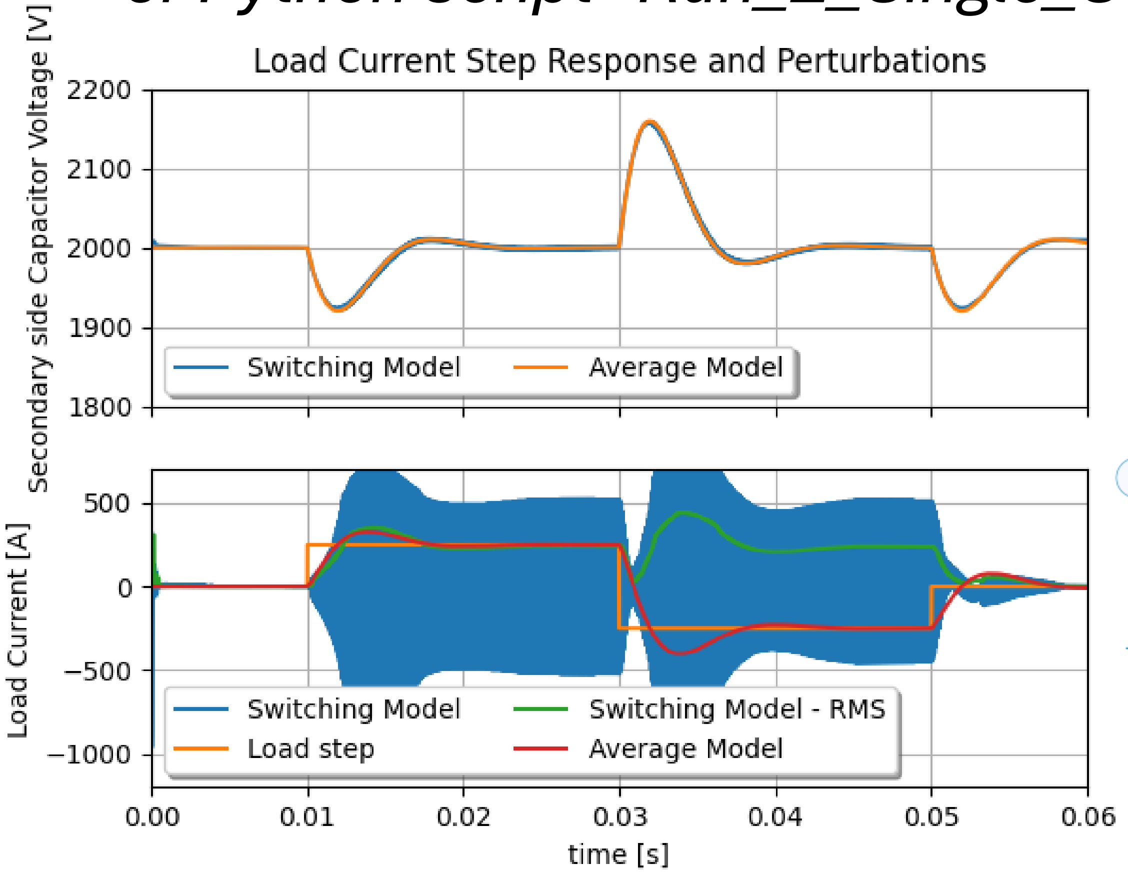
ELEMENTARY DAB-BASED SST CELL · CURRENT LIMITATION

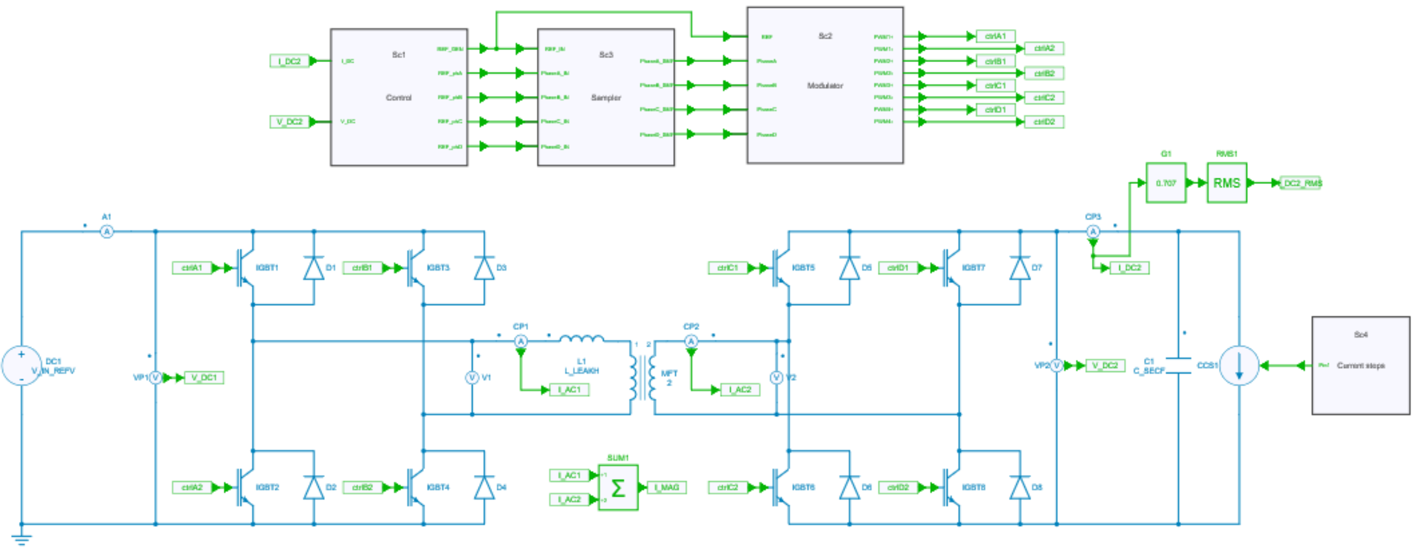
- With a current limiter, an effect on the voltage control output is observed and possibly affects overshoots and dynamics
- One can also use this limiter for short circuit limitation feature



Comparison between switching and average model

- *Run model “Semiconductor level Blocks - Voltage Control” from file “SST_Switched_Model.jsimba” or Python script “Run_2_Single_SST_comparison”*

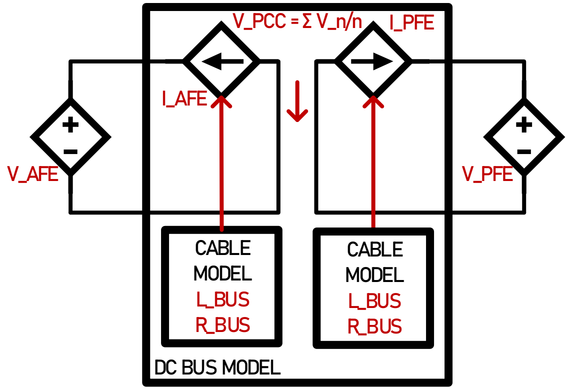




DC BUS FOR CONNECTING TWO VOLTAGE SOURCES

- Connecting voltage sources can only be done with current sources
- In this case we use a cable model with resistance and inductor for generating the current
- The Voltage as seen at the Point of Common Coupling is the pondered mean value between all applied voltages

$$V_{PCC} = \frac{V_{AFE}+V_{PFE}}{2}$$
$$I_{AFE} = \frac{1}{L_{DCBUS}} \int (V_{PCC} - V_{AFE} - R_{DCBUS} I_{AFE})$$
$$I_{PFE} = \frac{1}{L_{DCBUS}} \int (V_{PCC} - V_{PFE} - R_{DCBUS} I_{PFE})$$



Active Front End and Battery Energy Storage System

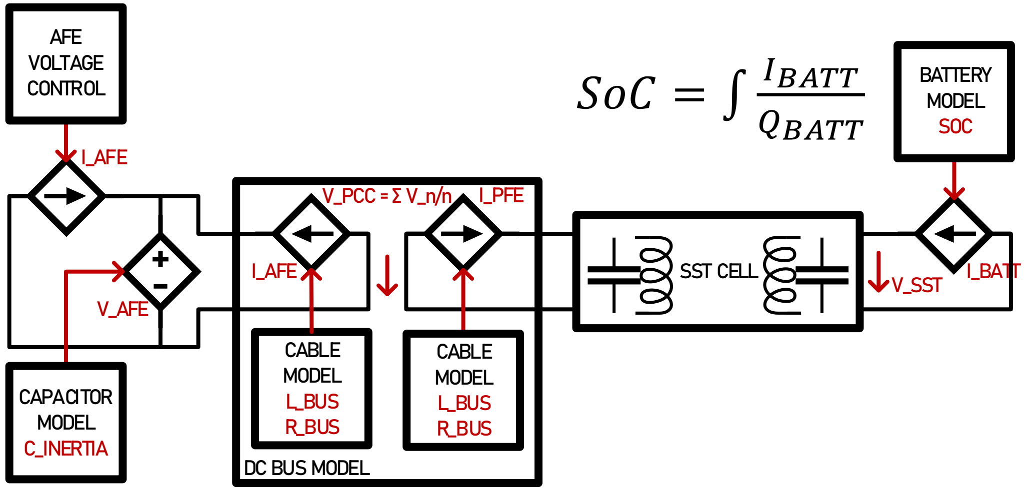
- The active front end is a current source feeding a voltage source
- The dynamics are related to a PI controller that reflects real hardware
- The voltage source implements a so-called "synthetic inertia" modelled through its capacitive value

$$V_{AFE} = \frac{1}{C_{AFE}} \int (I_{FEED} + I_{AFE})$$

$$I_{FEED} = K_P(V_{AFE} - V_{DC}) + K_I \int (V_{AFE} - V_{DC})$$

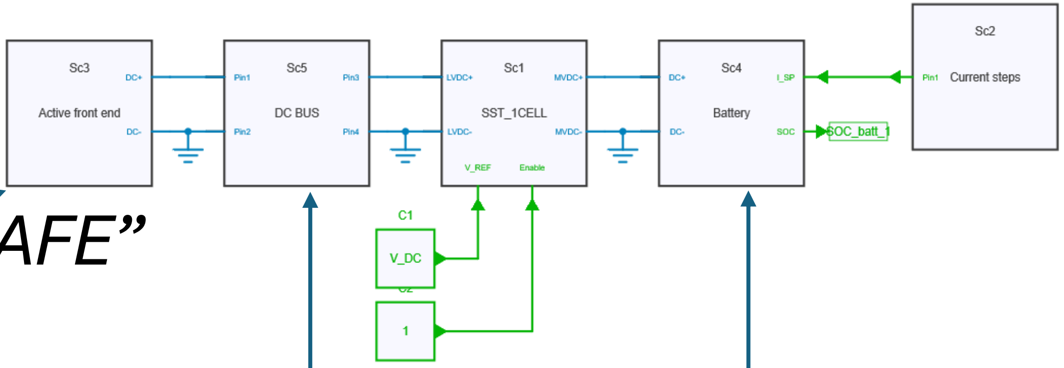
- The Battery model implements the idea of State of Charge connected to a current source.
- The Battery can only operate between 10 and 90%

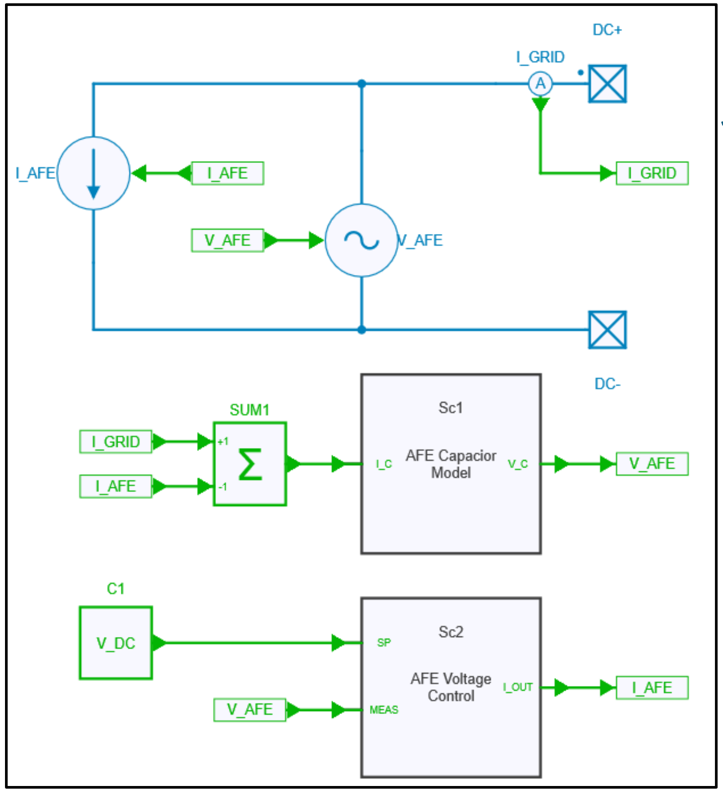
$$SoC = \int \frac{I_{BATT}}{Q_{BATT}}$$

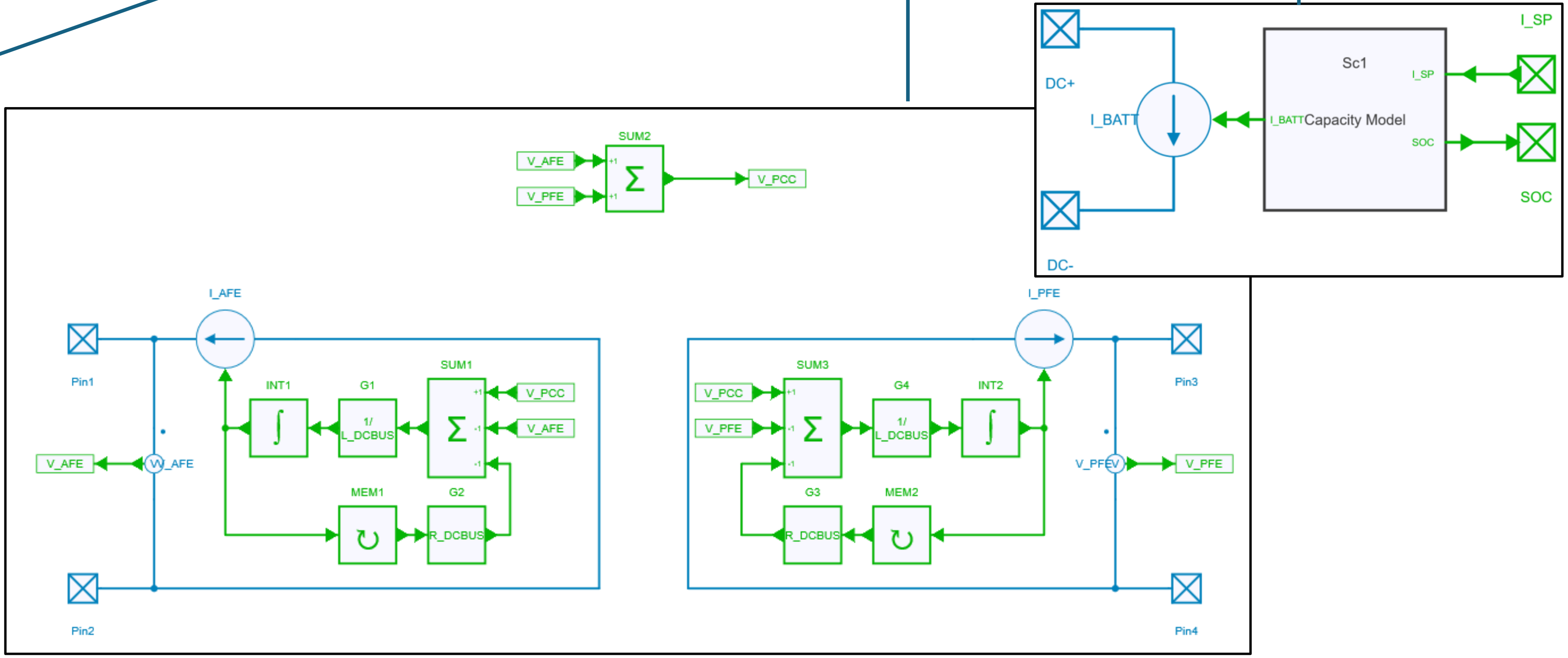


Active Front End and Battery Energy Storage System

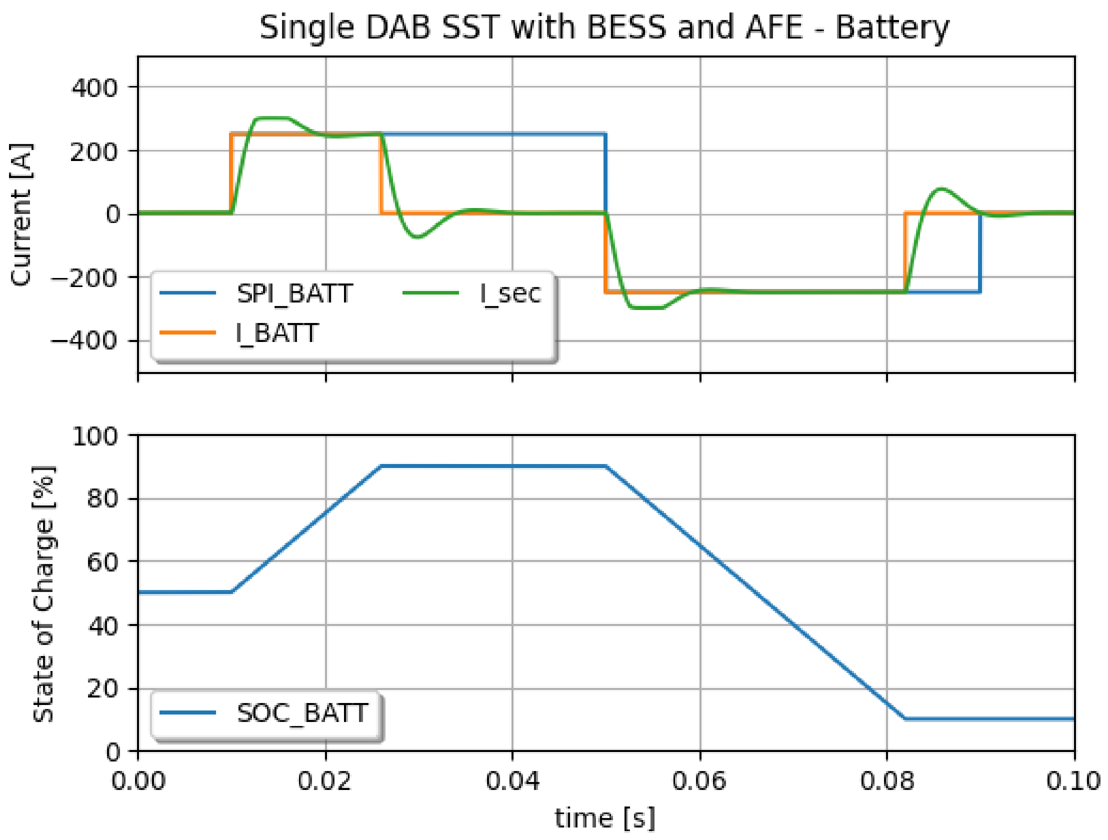
- *Run model "3 Single SST with BESS and AFE" from file "SST_DCMicroGrid_Models.jsimba" or Python script "Run_3_Single_STT_Battery_AFE"*

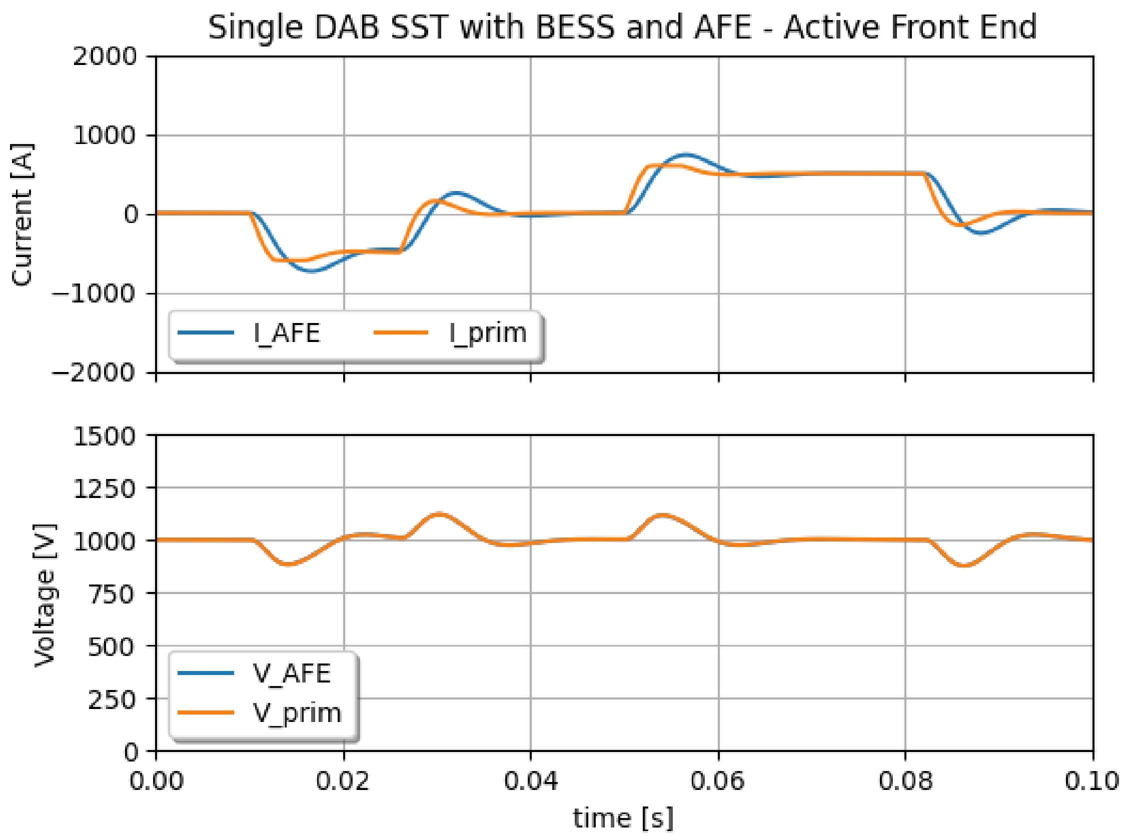






Active Front End and Battery Energy Storage System





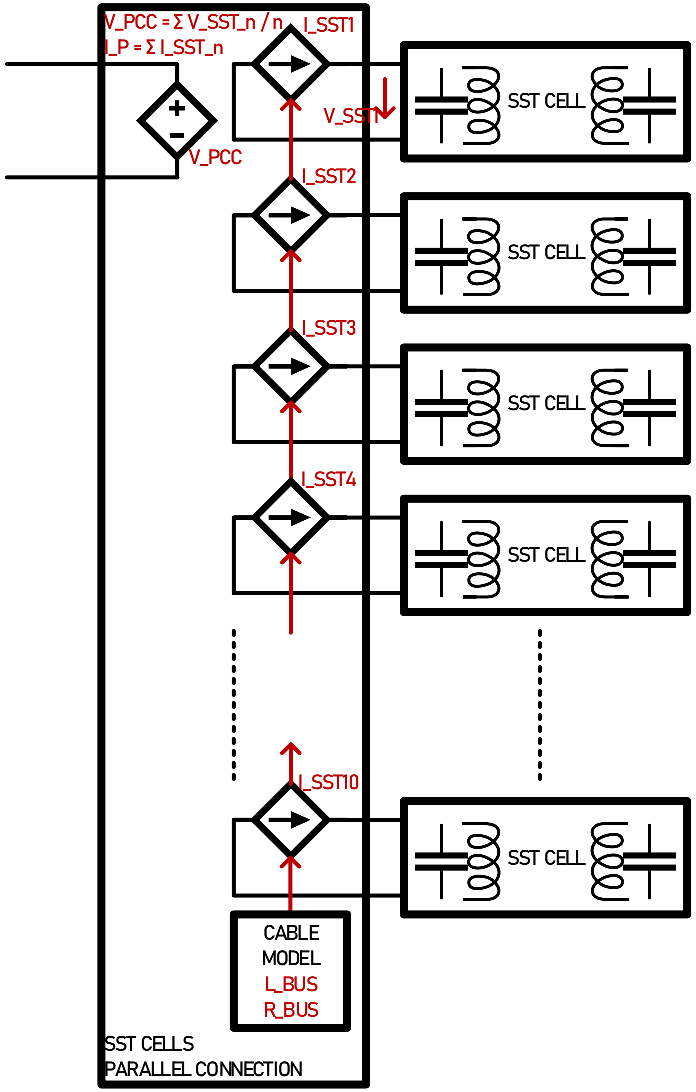
Series/Parallel connection of SST cells

- Connecting voltage source in series is not an issue
- Connecting voltage sources in parallel goes through current sources that implement a cable model
- The output voltage is a voltage source taking the mean value between them all

$$V_{PCC} = \sum \frac{V_{PRIM\_N}}{N}$$

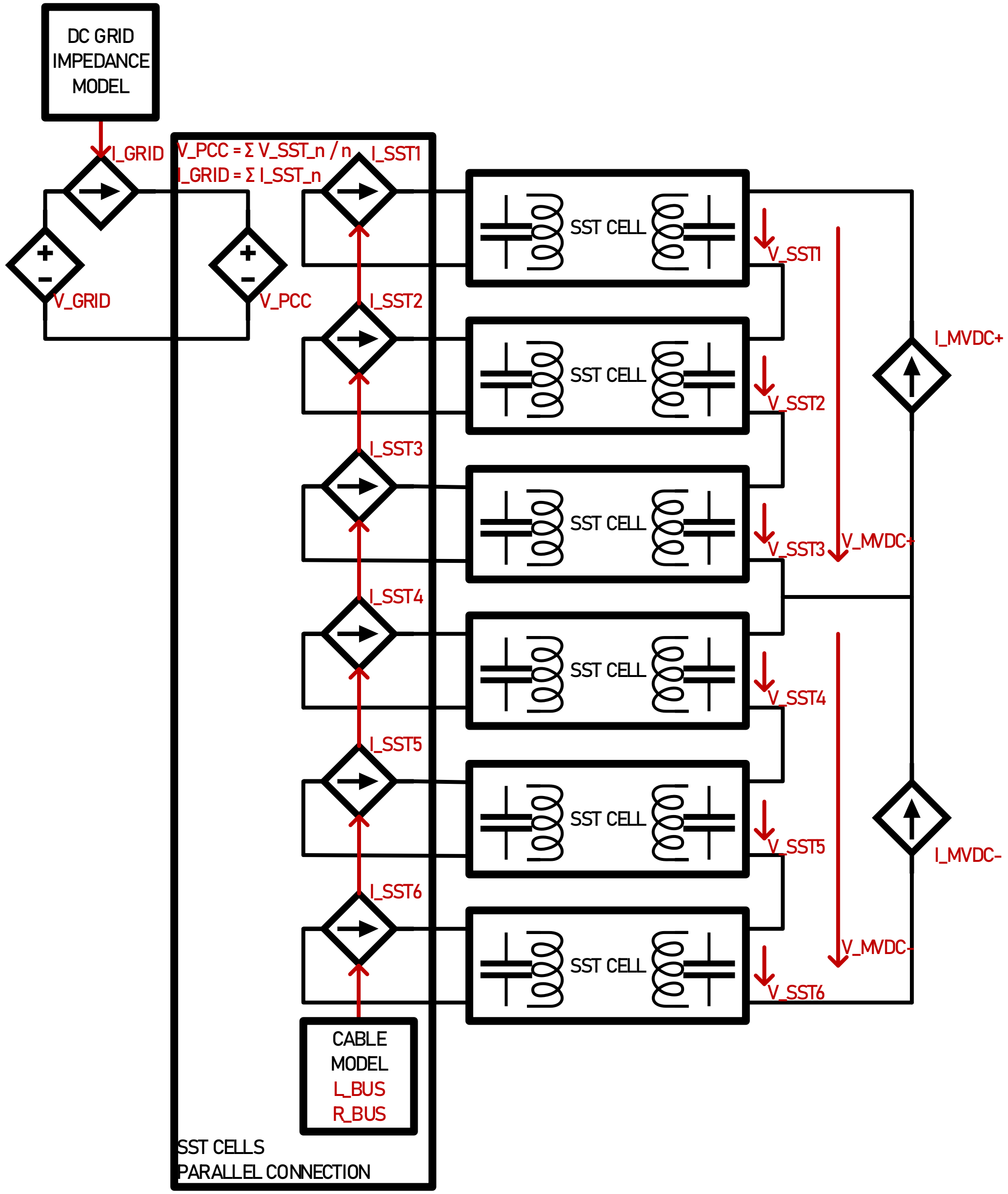
$$I_{GRID} = \sum I_{PRIM\_N}$$

$$I_{SST\_N} = \frac{1}{L_{DCBUS}} \int (V_{PCC} - V_{PRIM\_N} - R_{DCBUS} I_{PRIM\_N})$$



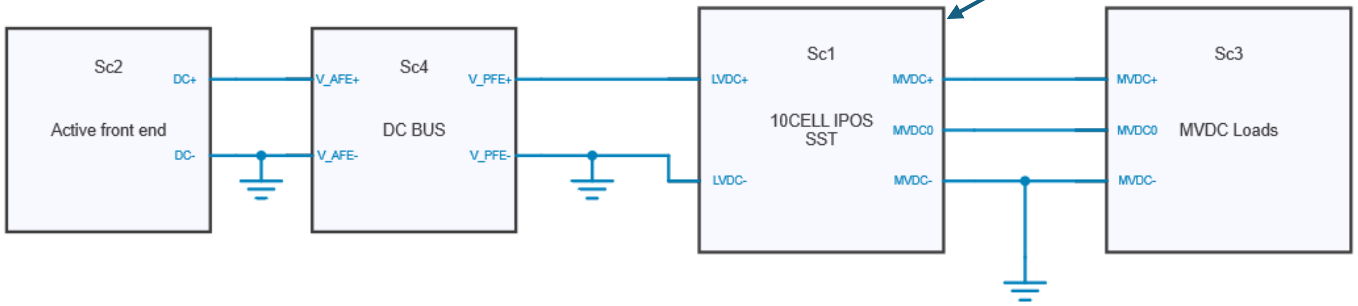
MULTICELL IPOS CONFIGURATION

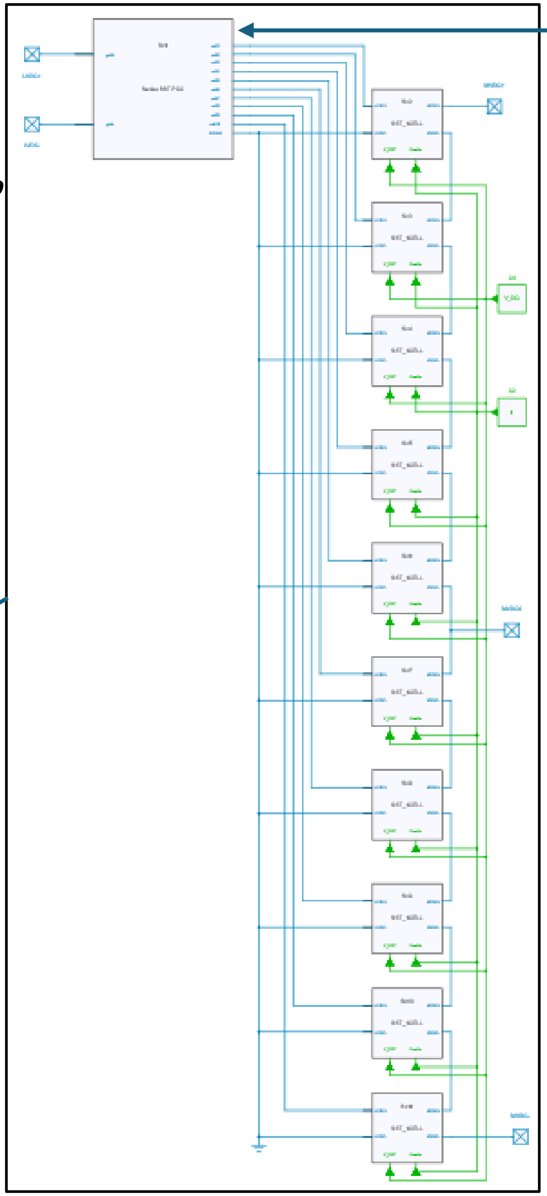
- The IPOS stands for
  Input Parallel Output Series
- The Output (right hand side) is connected to a three port MVDC with two current sources which may be asymmetric
- Voltages are controlled separately on each SST Cell secondary side
- LVDC Input (left hand side) implements a grid impedance model

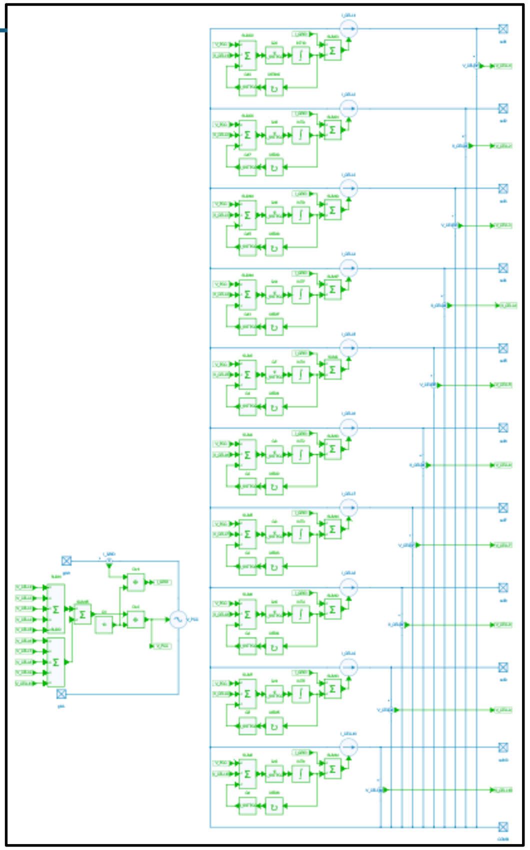


MULTICELL IPOS CONFIGURATION

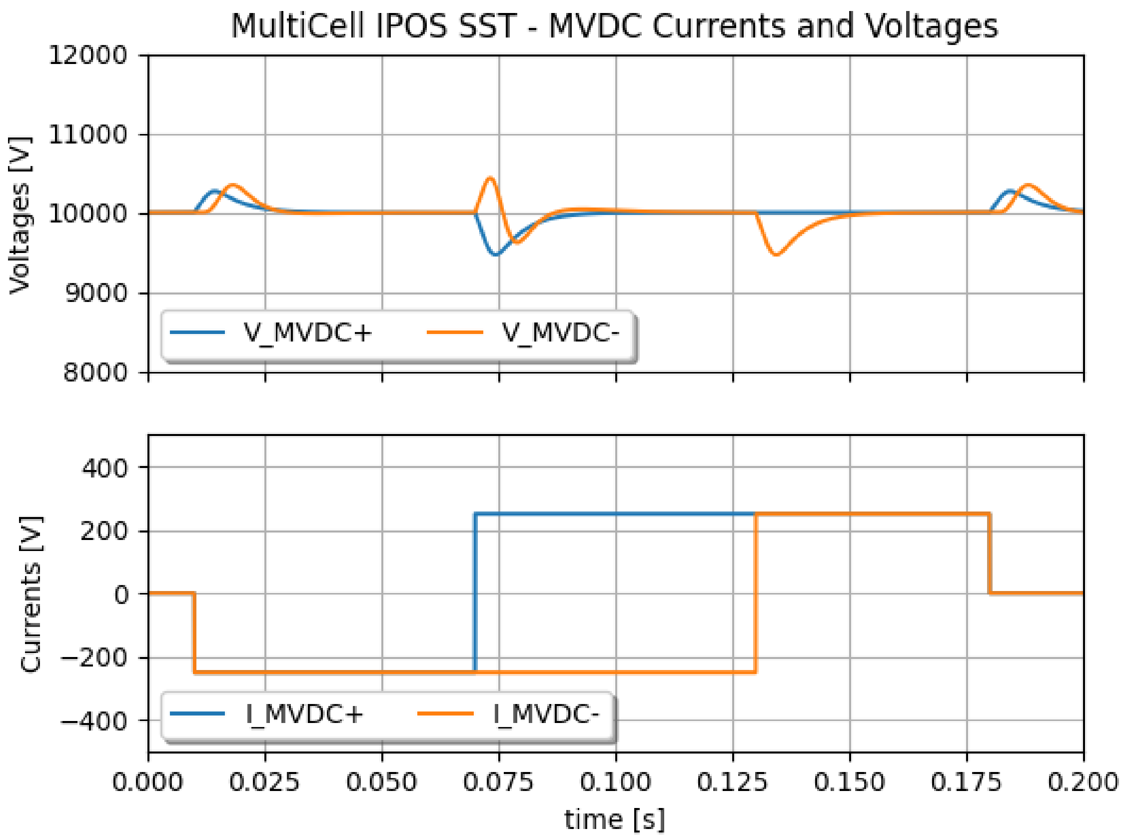
- *Run model "4 MultiCell IPOS SST" from file "SST_DCMicroGrid_Models.jsimba" or Python script "Run_4_10cell_IPOS"*

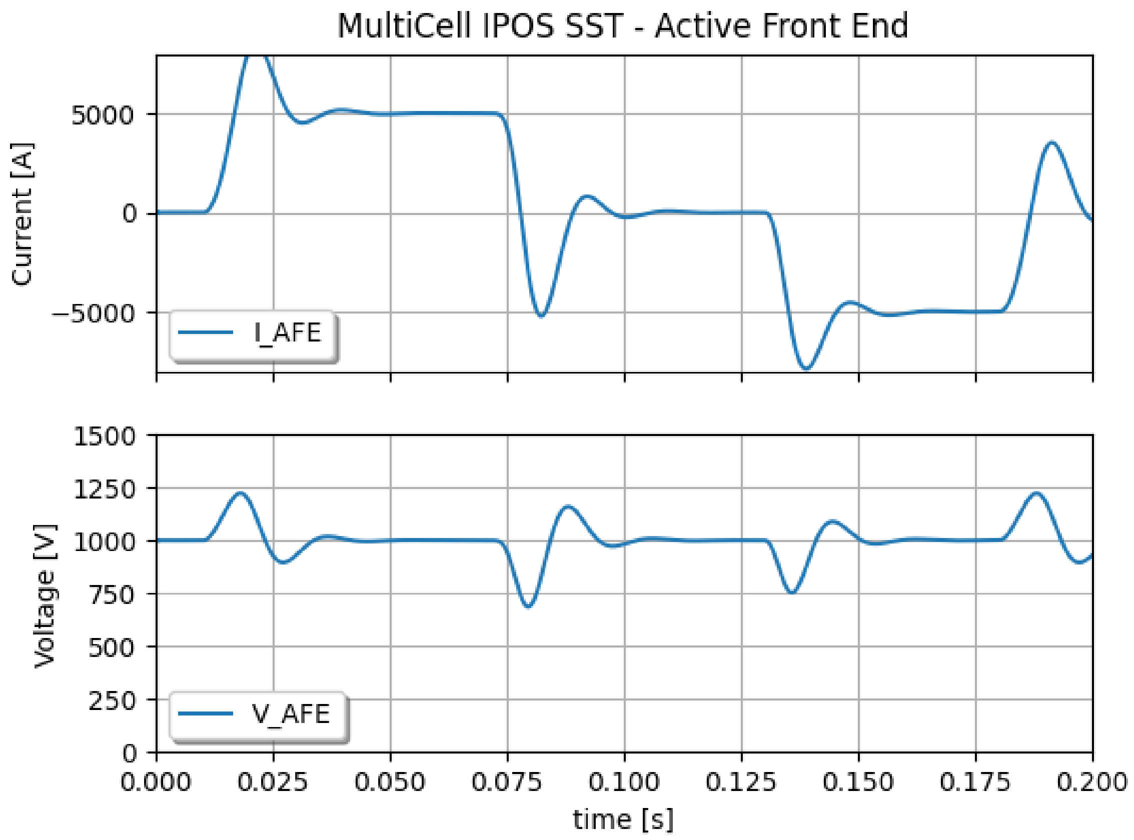




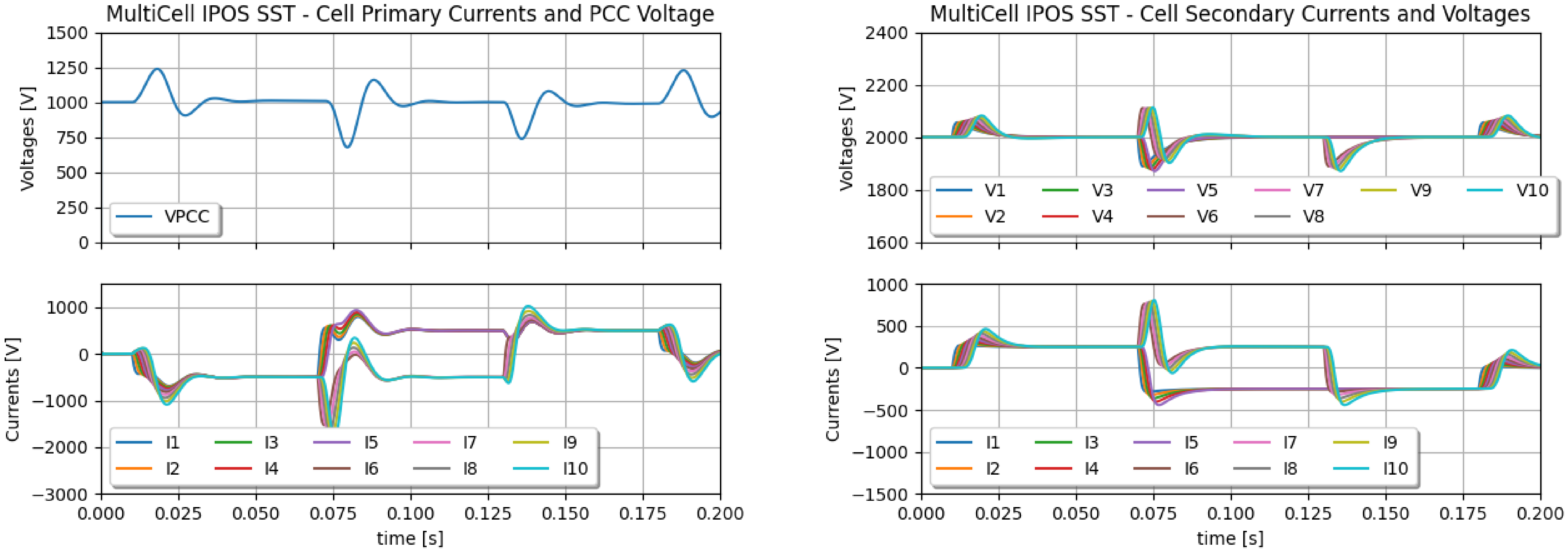


MULTICELL IPOS CONFIGURATION



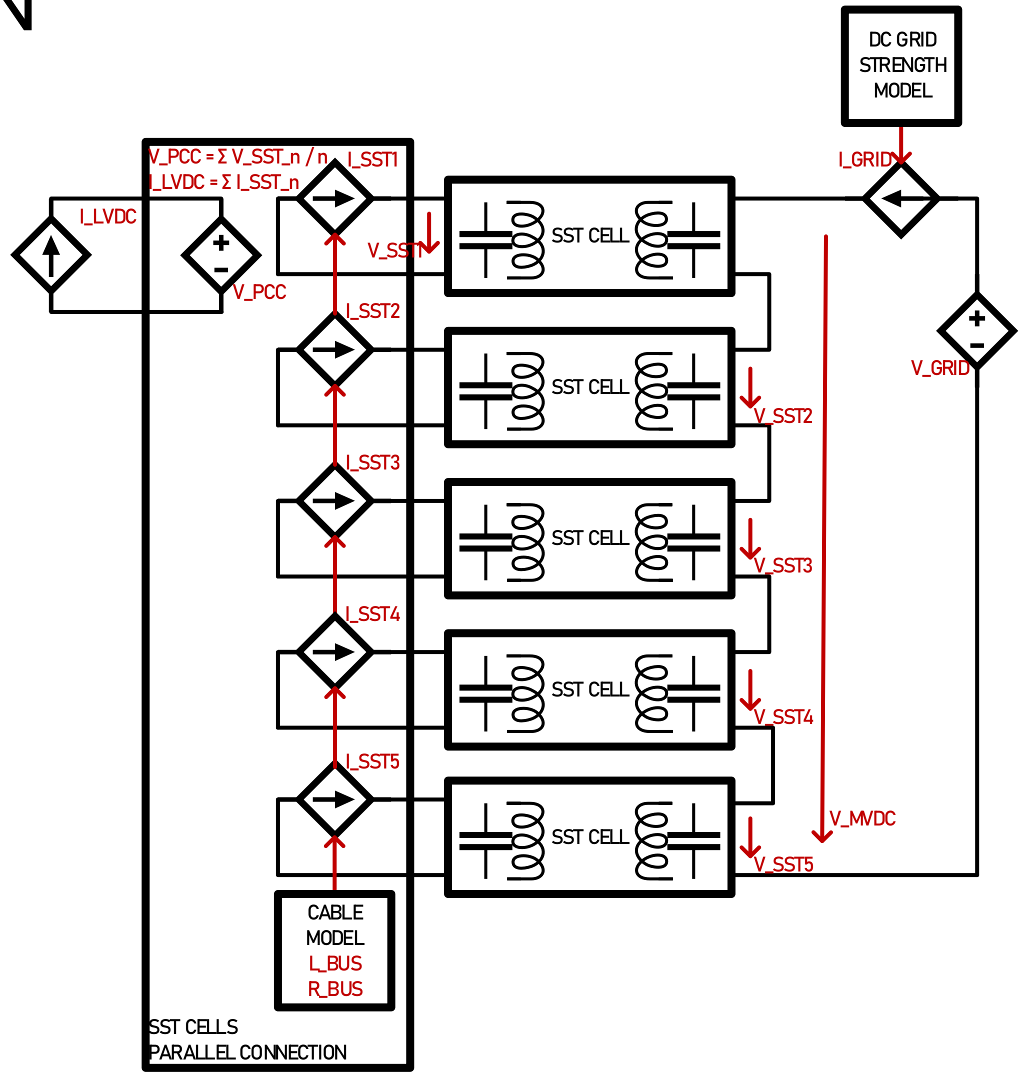


MULTICELL IPOS CONFIGURATION



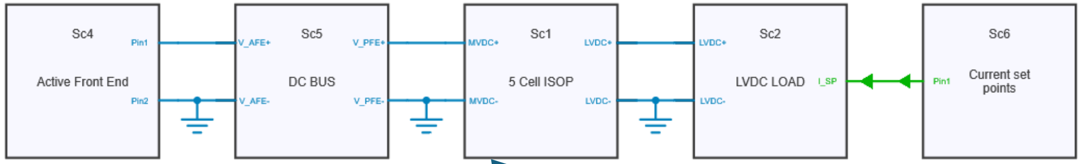
MULTICELL ISOP CONFIGURATION

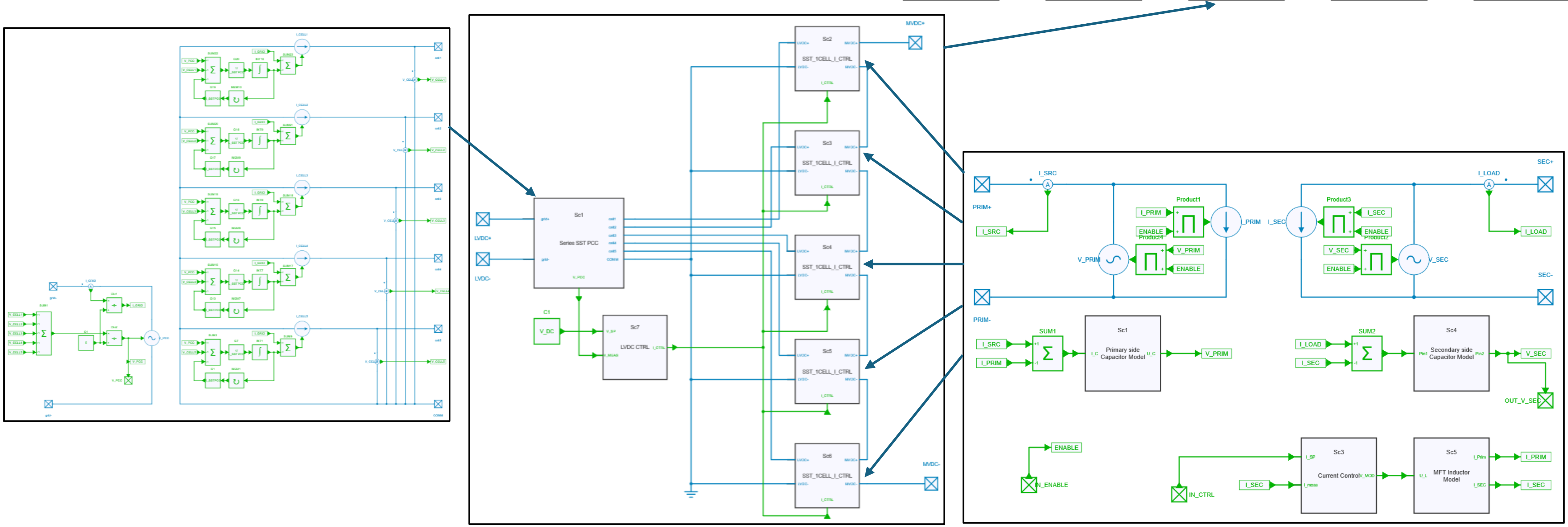
- The ISOP stands for Input Series Output Parallel
- The Output (left hand side) is connected to an LVDC current source
- Voltage is controlled on primary side by a single controller, each cell takes same current set point for ensuring secondary side voltage balance
- MVDC Input (right hand side) implements a grid impedance model



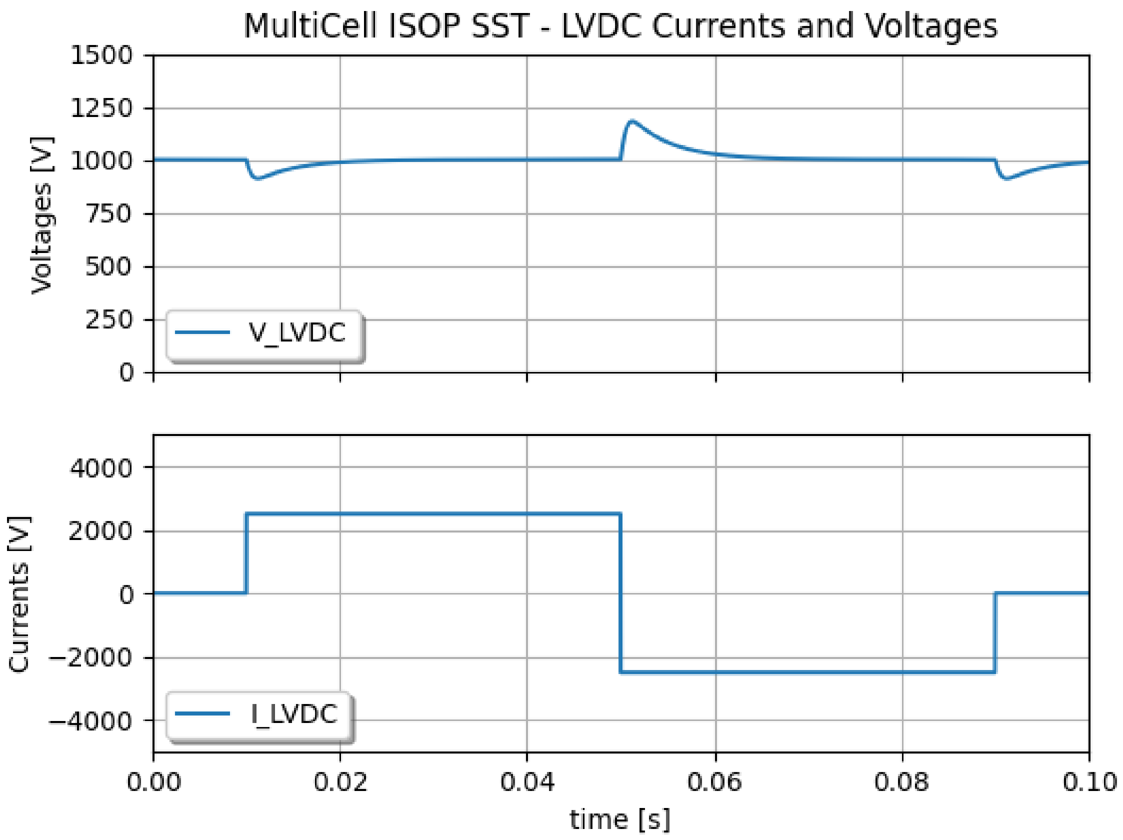
MULTICELL ISOP CONFIGURATION

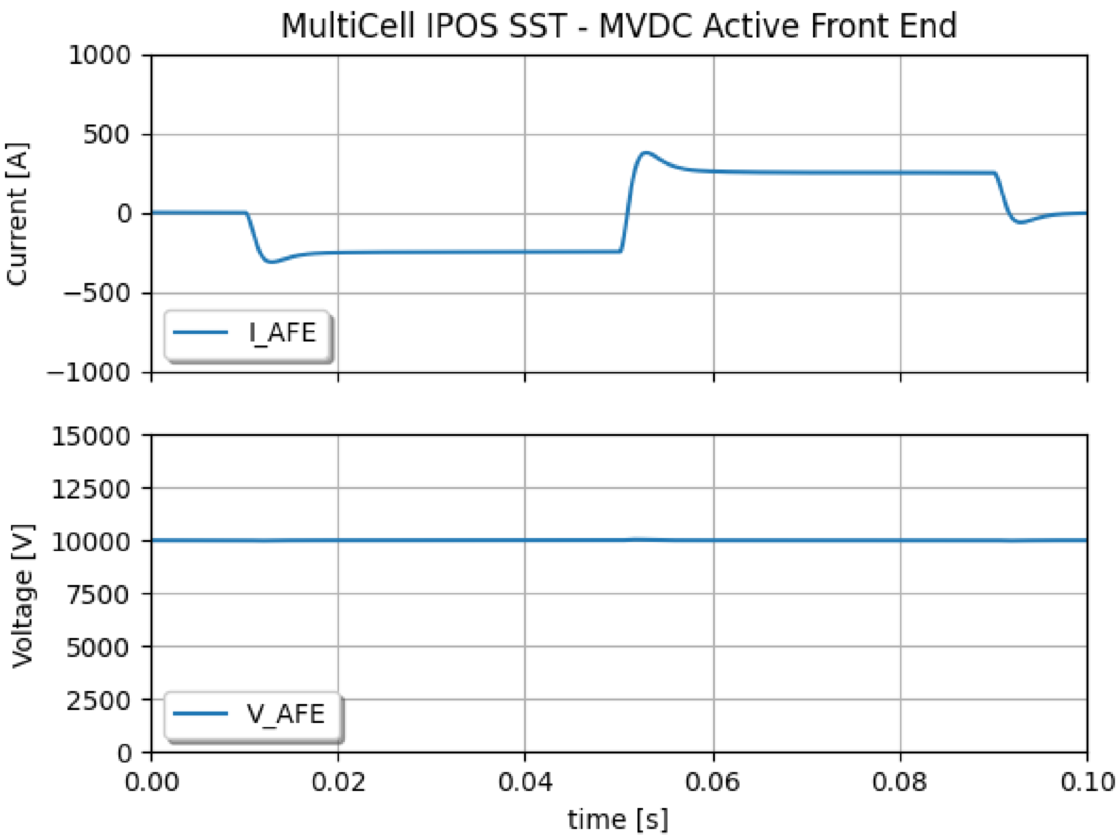
- *Run model “5 MultiCell ISOP SST” from file “SST_DCMicroGrid_Models.jsimba” or Python script “Run_5_5cell_ISOP”*



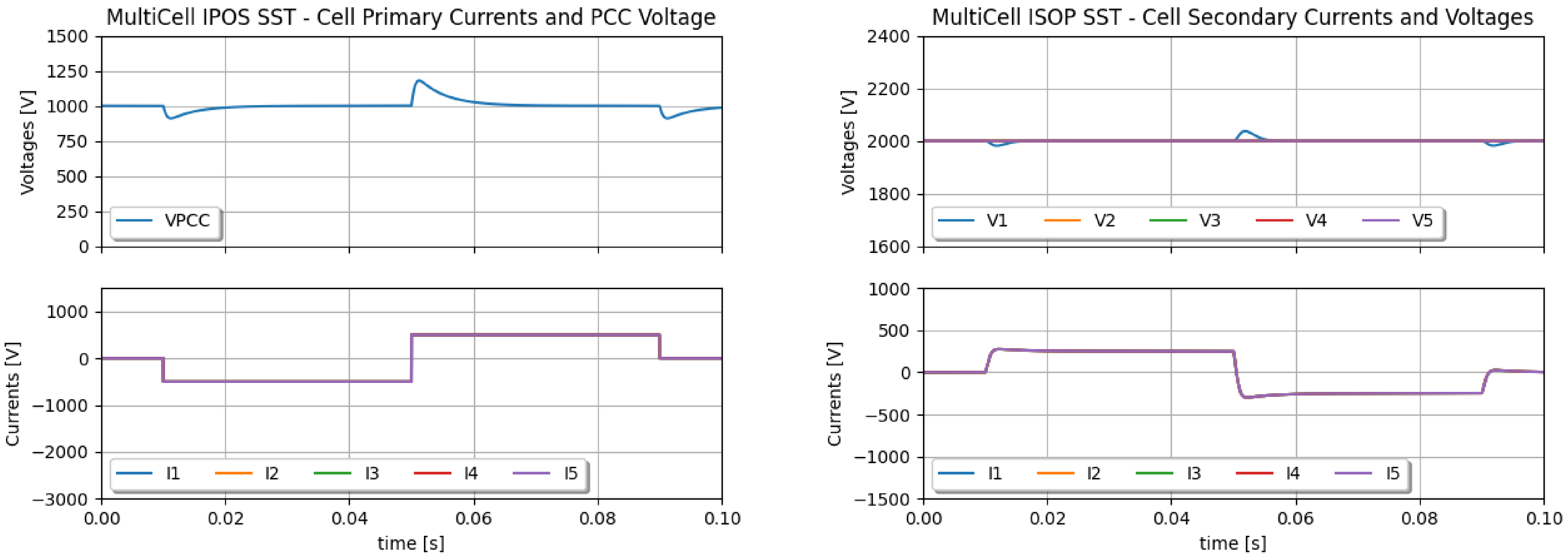


MULTICELL ISOP CONFIGURATION





MULTICELL ISOP CONFIGURATION



Model of an MVDC microgrid

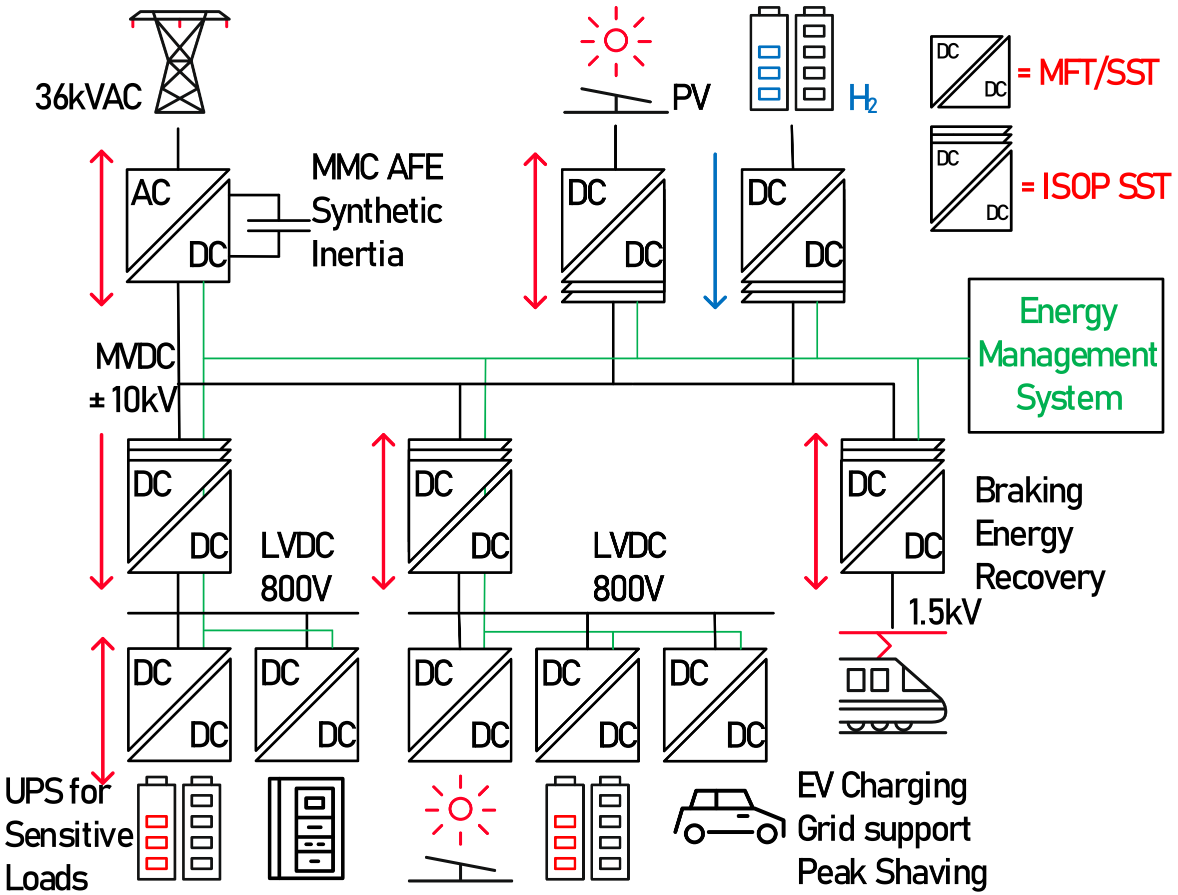
- Full Microgrid with an MVDC backbone
- Run scenario for MVDC power system
- Continuous run approach with Python
- Conclusions and prospect

References:

Daniel Siemaszko and Mauro Carpita, "**Continuous Time Simulation and System-Level Model of a MVDC Distribution Grid Including SST and MMC-Based AFE**," MDPI special issue on Multi-Level Power Converter Systems, 2024

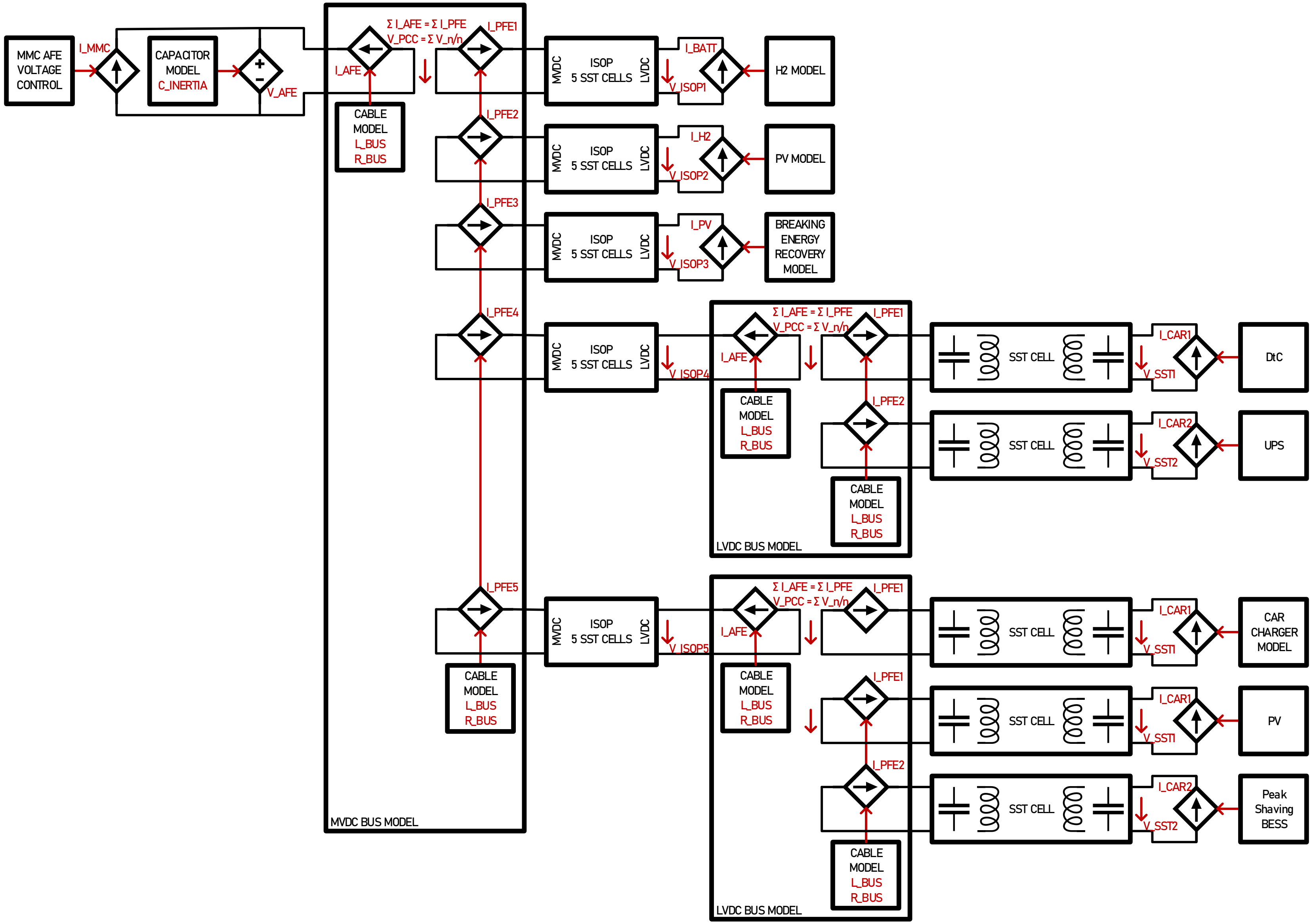
FULL MICROGRID WITH AN MVDC BACKBONE

- In the following example, a full microgrid is operated with an MVDC backbone
- The MVDC backbone is in this case controlled by one active front end connected to the grid
- It features sources, loads, storage, and two connection to LVDC buses
- LVDC links features local battery storage for either UPS function or peak shaving



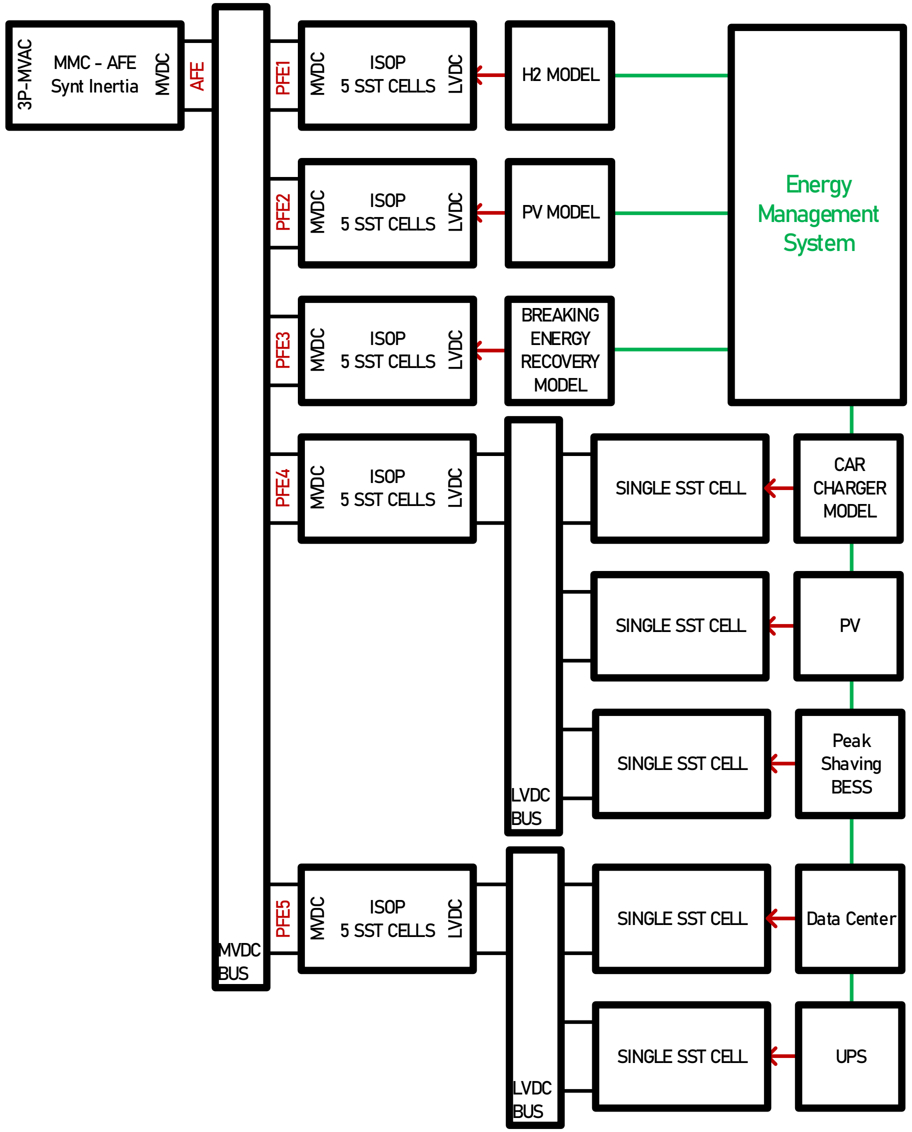
Full Microgrid with an MVDC backbone

- The model is built with all previously presented elementary blocks
- About 30 SSTs are independently running in this model



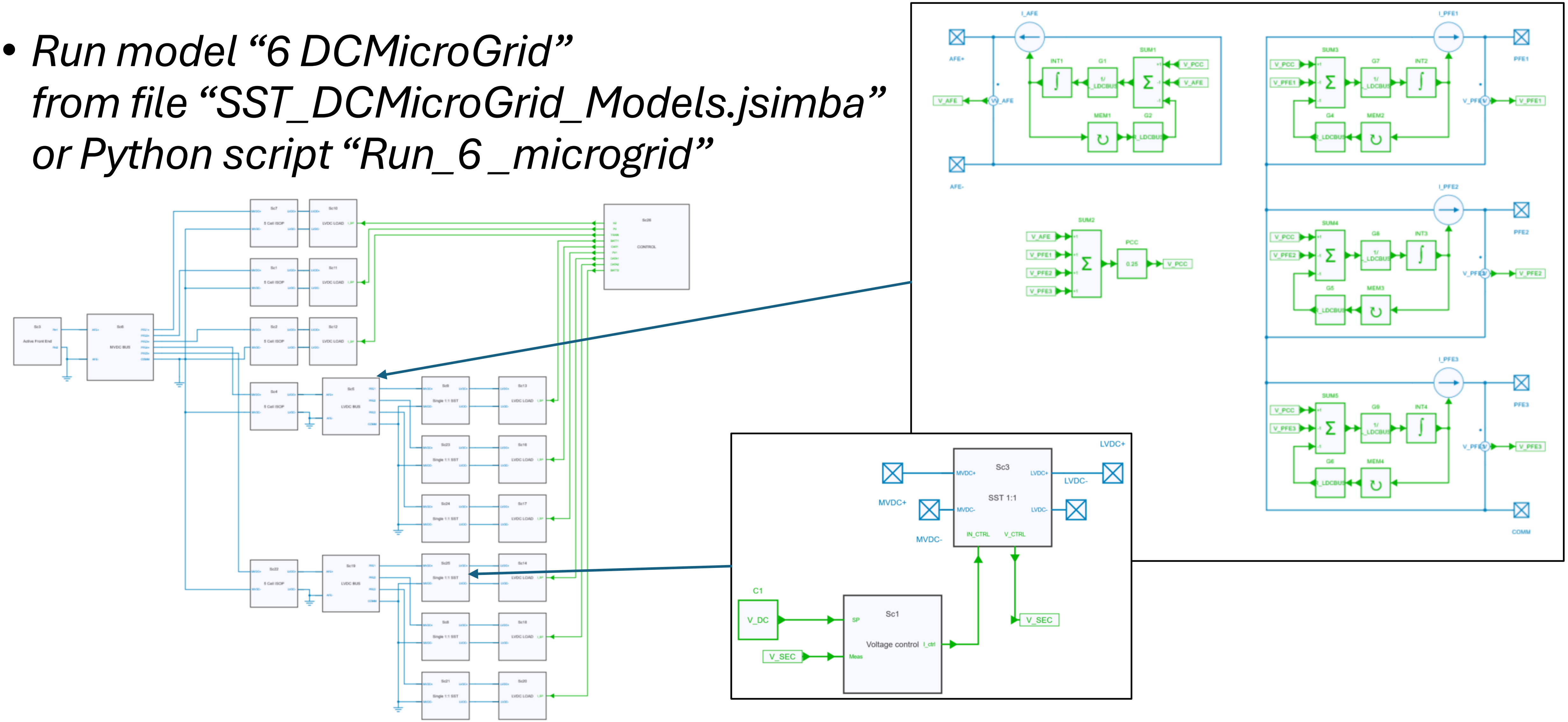
Run scenario for MVDC power system

- The set points for sources, loads, storage systems are centralized in the Energy Management System
- A full day is simulated with PV generation (including drops due to clouds), grid support from Hydrogen, breaking energy recovery from trains
- Local LVDC busses features battery storage for local UPS (typically Data Center) and local Peak Shaving (typically residential area with PV and car charging)



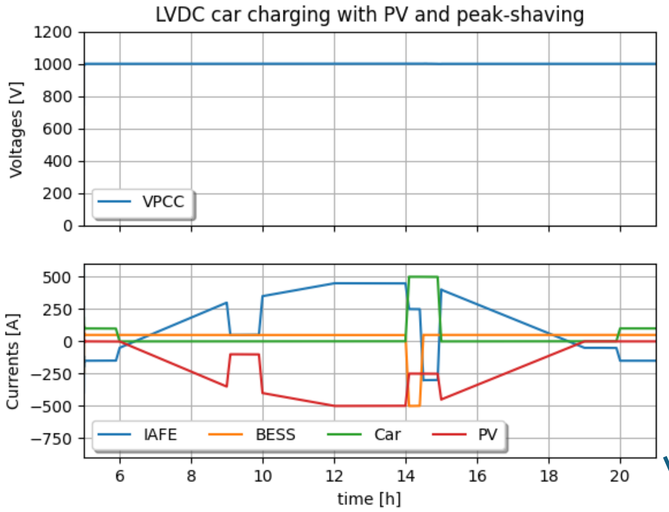
RUN SCENARIO FOR MVDC POWER SYSTEM

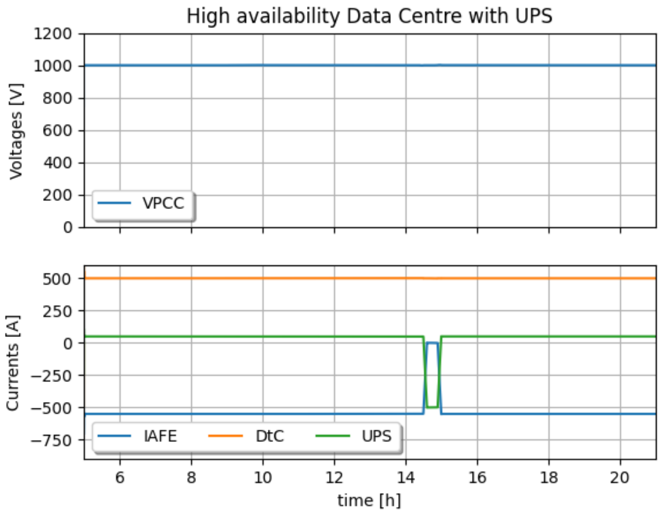
- *Run model "6 DCMicroGrid" from file "SST_DCMicroGrid_Models.jsimba" or Python script "Run_6_microgrid"*

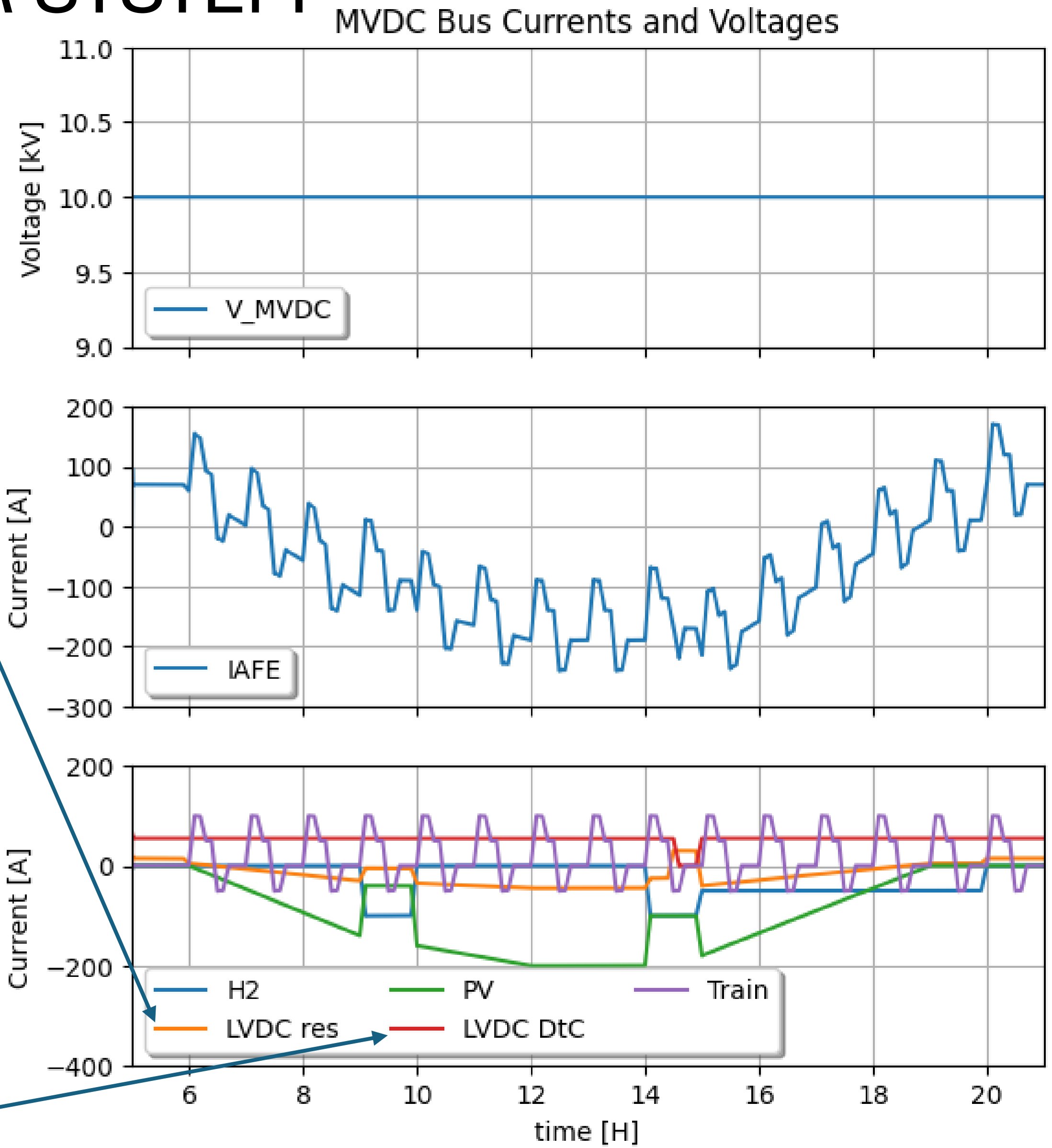


RUN SCENARIO FOR MVDC POWER SYSTEM

- At some point, a car fast charge is requested, but a PV drop occurs
- Power is taken from utility grid but AFE is not enough for all loads
- DtC temporarily switches to UPS for saving the day

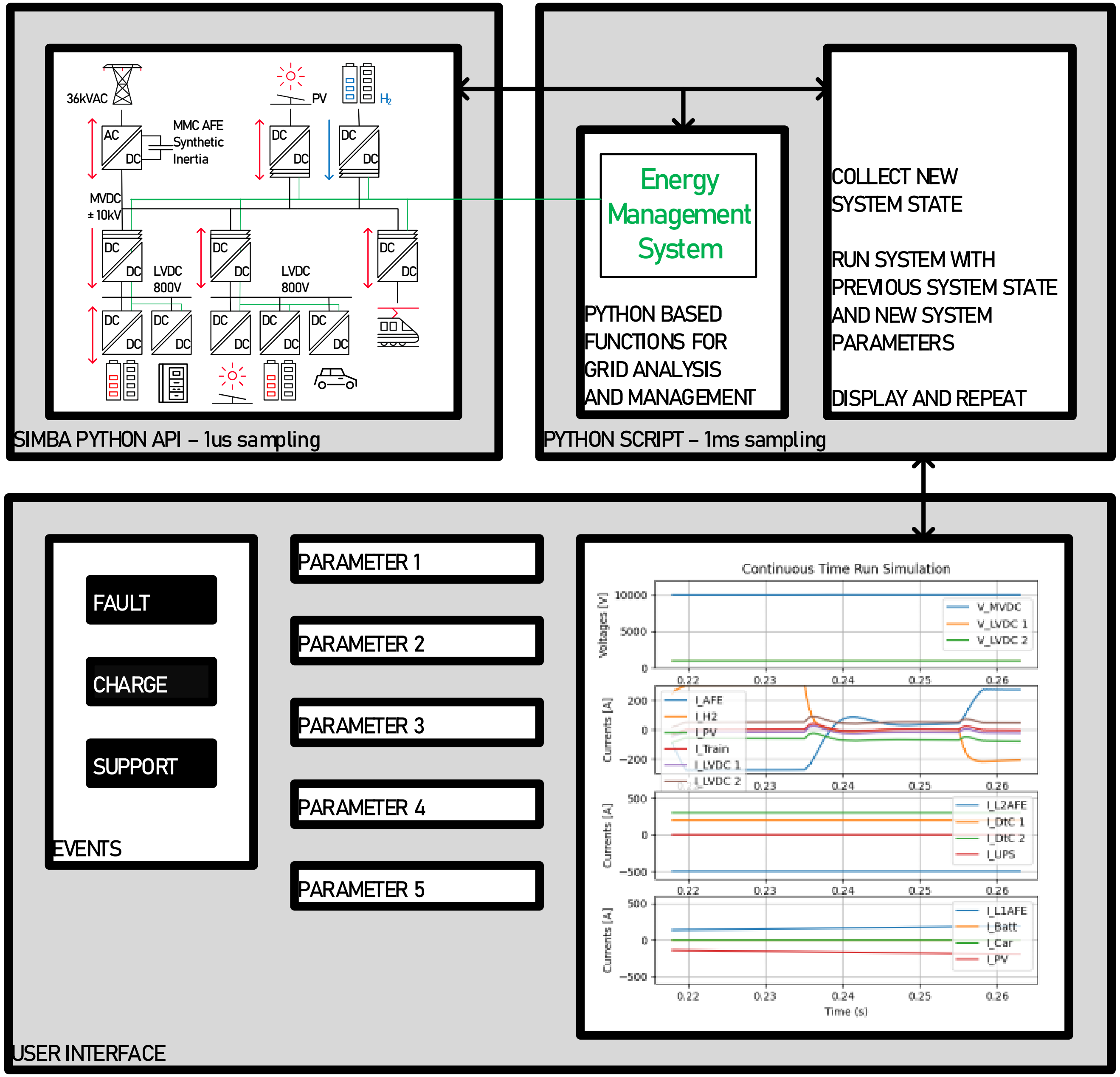






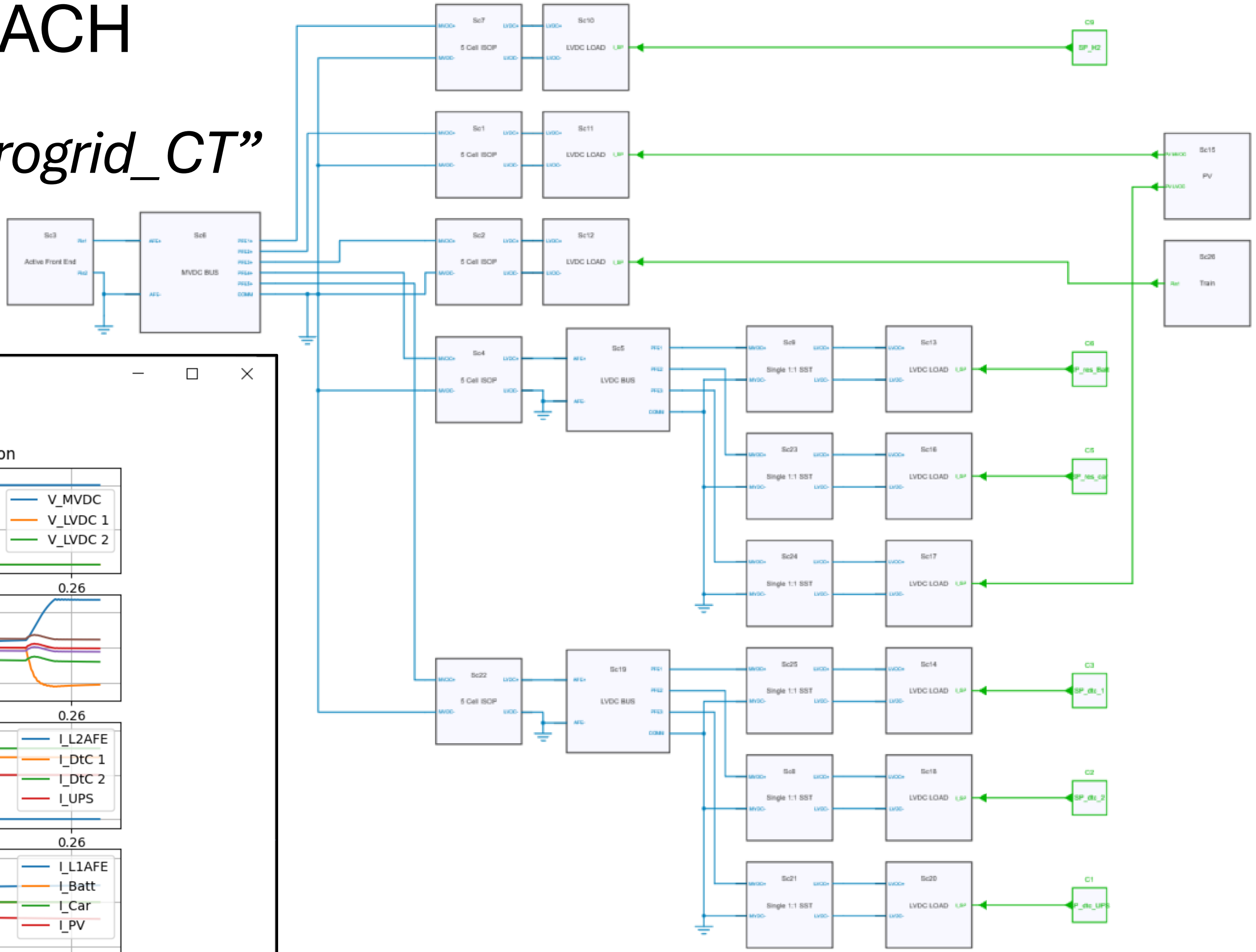
CONTINUOUS RUN APPROACH

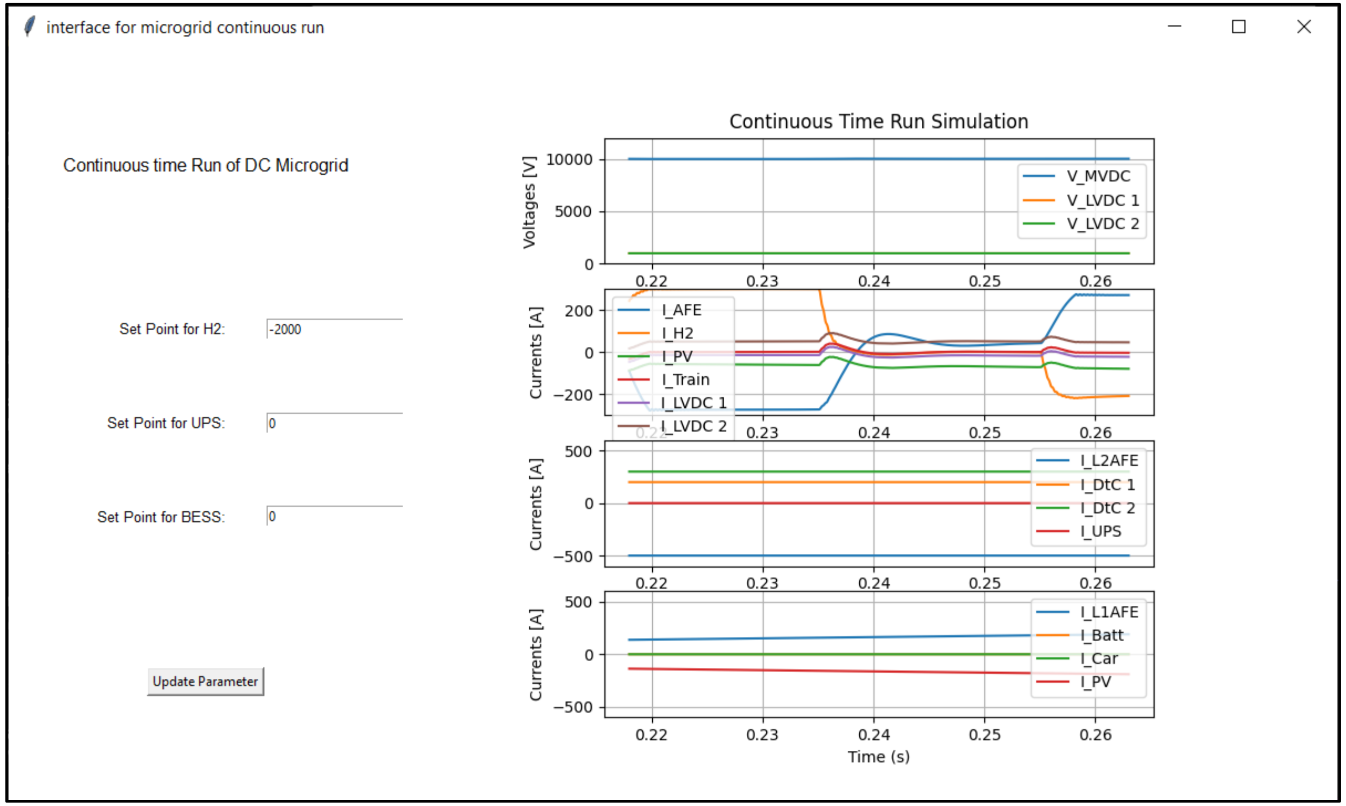
- The Simba model
  - Running at pseudo 1us sampling rate
  - Snapshot principle
- A Python user interface
  - Plot visualization
  - Parameters / set points settings
  - Possible alarms
- A Phython script
  - Calling the Simba Python API
  - Running at pseudo 1ms sampling rate
  - Implementing possible automated energy management features



CONTINUOUS RUN APPROACH

- *Run Python script “Run_7_microgrid_CT”*





Conclusions and Prospects

Contact me !

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Get the models !

https://github.com/PESC-CH/

- A didactic tool for simulating complex DC power systems with SST dynamics
- An approach to DC power system study
- Next steps would include
  - Modelling of droop controls for MVDC
  - Hardware and software protection schemes, short circuit current control
  - Possibly the implementation of any AI related Python toolboxes applicable to control power systems